

notebook

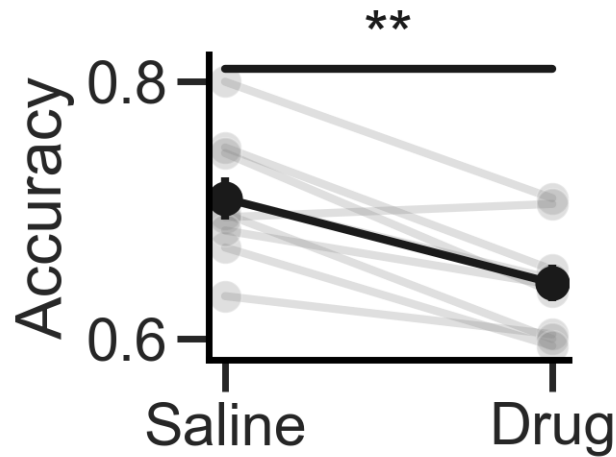
August 18, 2025

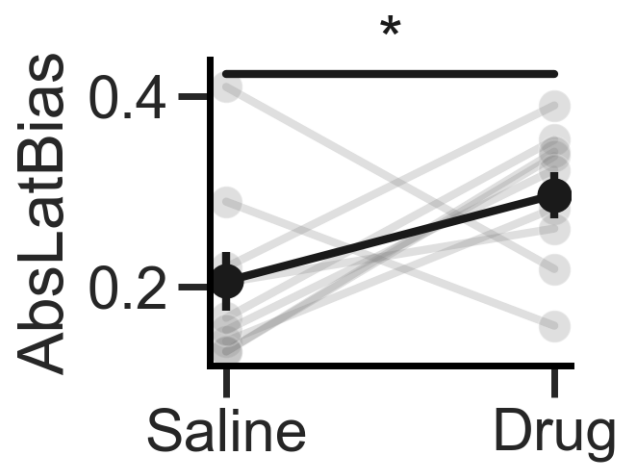
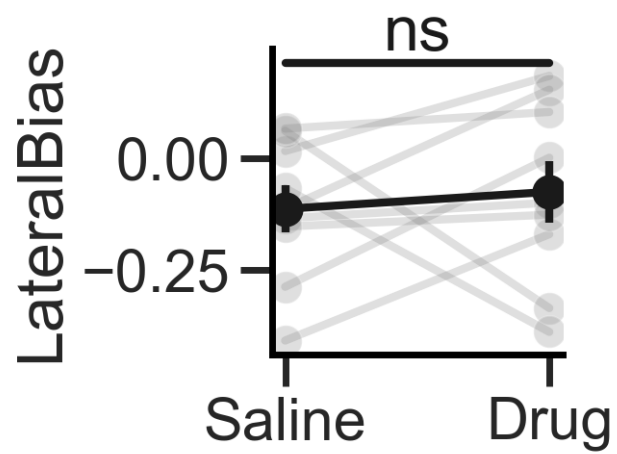
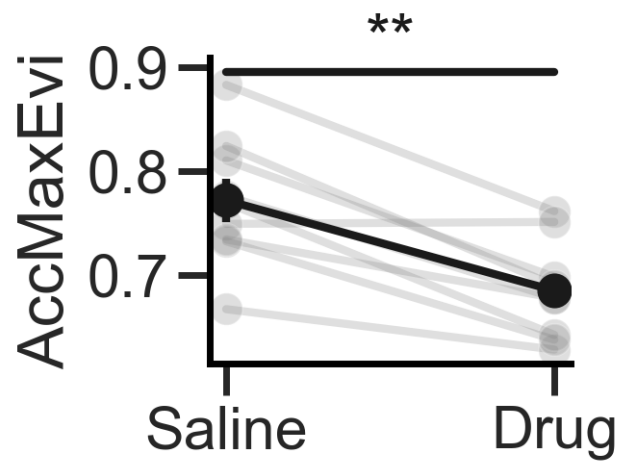
1 Analyses of drug data from batch #6

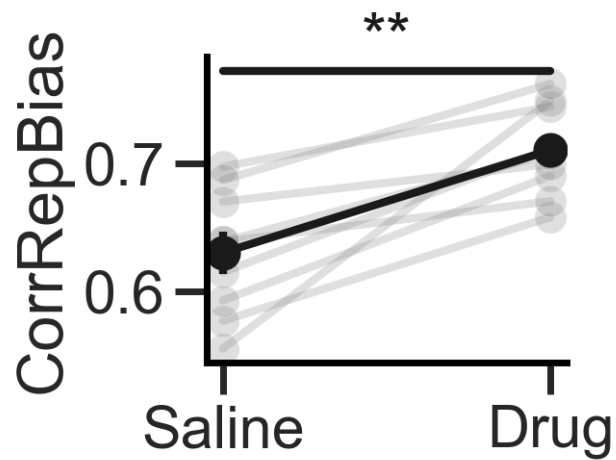
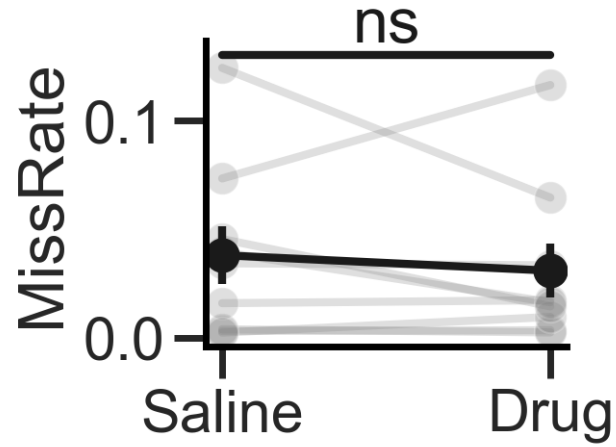
glue_animals_intersessions took 0.15 seconds to run

1.1 Behavioral metrics

Accuracy: t-statistic = 3.049, p-value = 0.008
AccMaxEvi: t-statistic = 3.333, p-value = 0.004
LateralBias: t-statistic = -0.441, p-value = 0.665
AbsLatBias: t-statistic = -2.308, p-value = 0.035
MissRate: t-statistic = 0.391, p-value = 0.701
CorrRepBias: t-statistic = -3.939, p-value = 0.001







1.2 Psychometrics

1.2.1 Probability choose RIGHT

Single animals

Animal 014

plot_pc took 1.22 seconds to run

plot_pc took 1.23 seconds to run

plot_pc_drug took 2.47 seconds to run

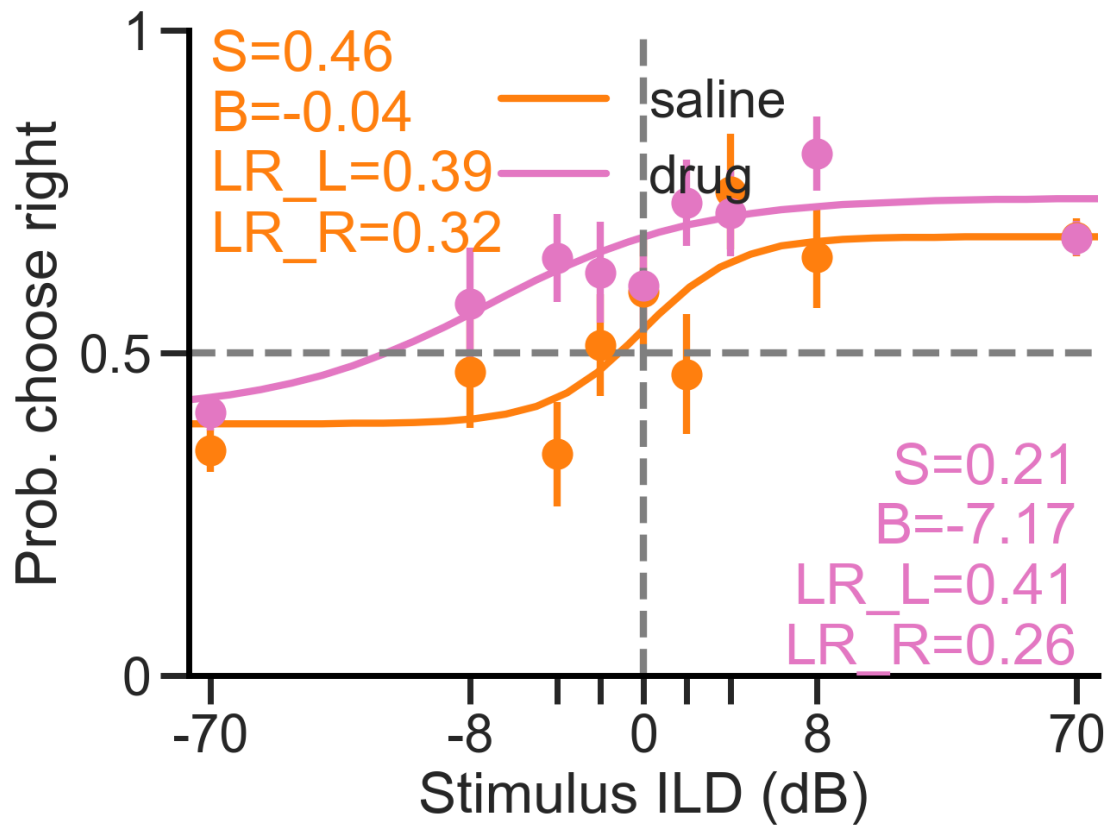
Animal 016

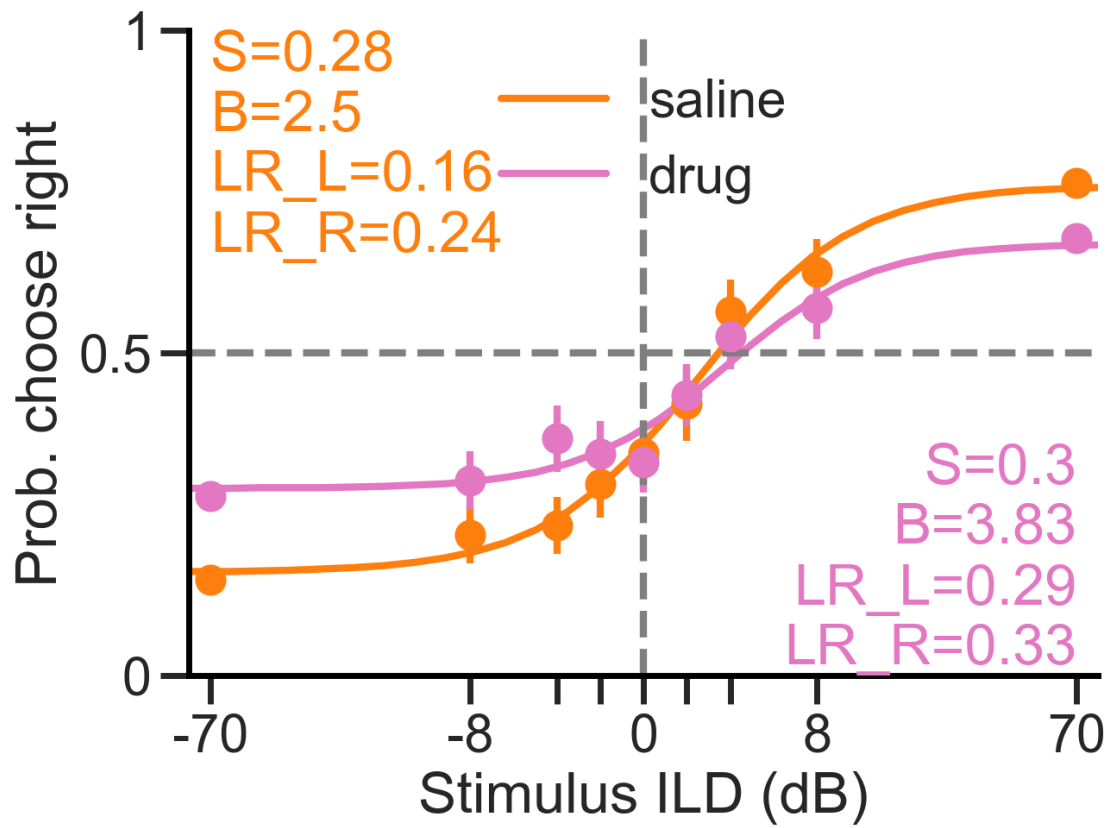
plot_pc took 1.27 seconds to run

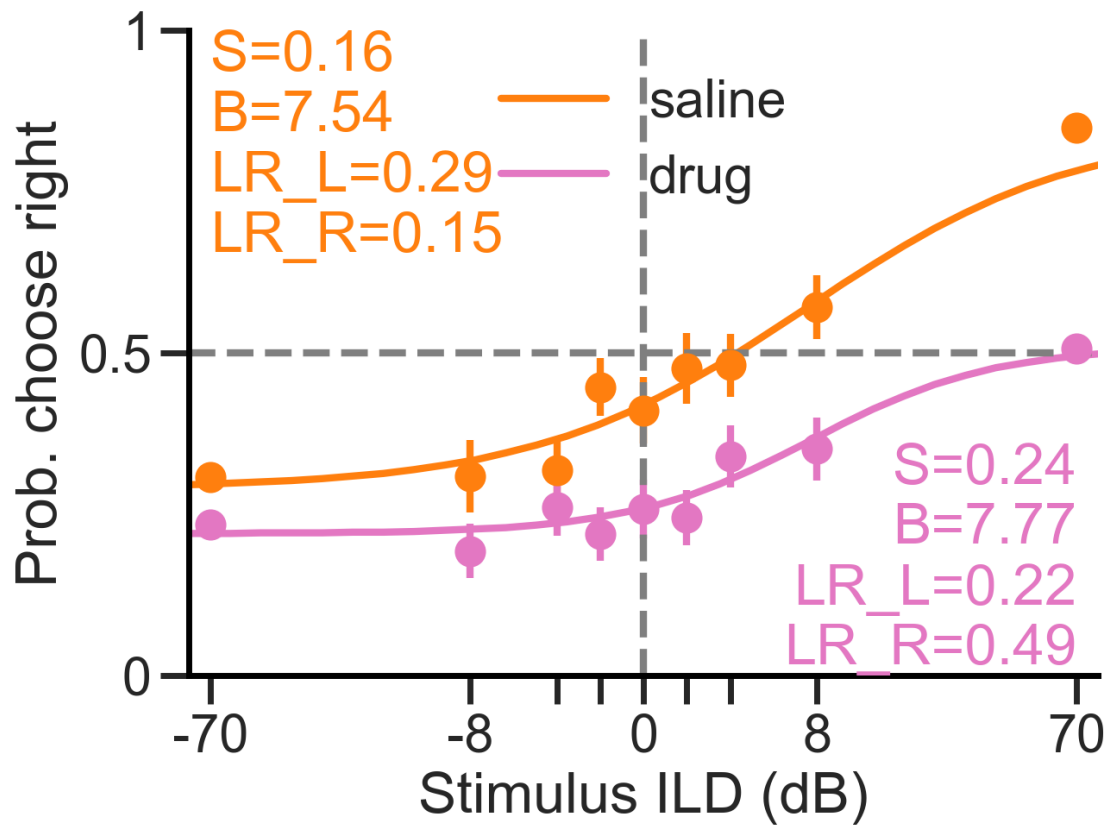
plot_pc took 1.31 seconds to run

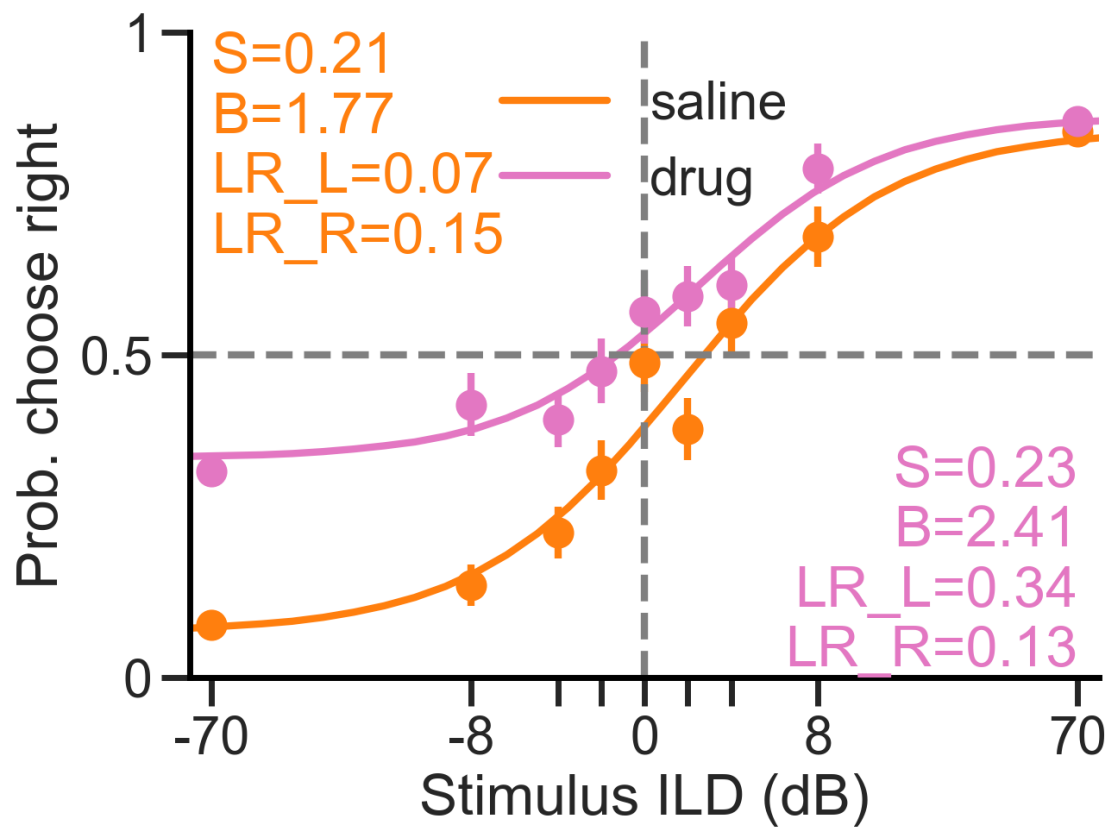
plot_pc_drug took 2.60 seconds to run

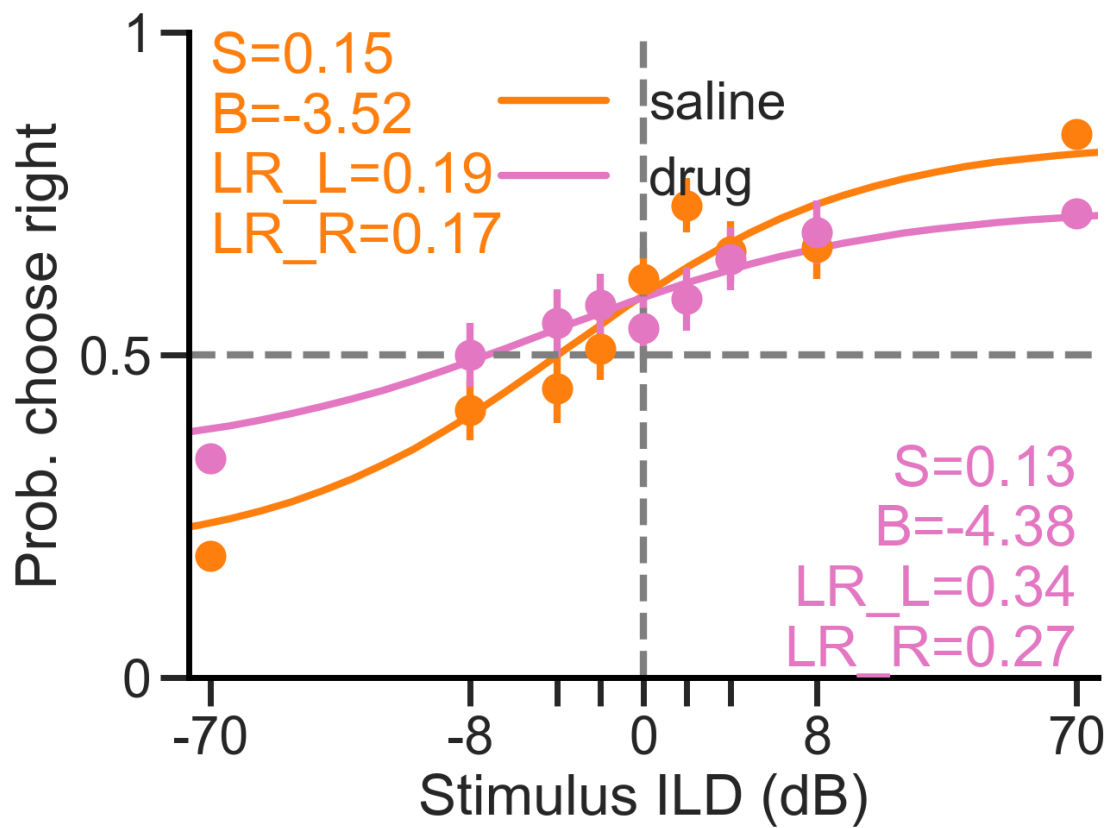
Animal 017
plot_pc took 1.44 seconds to run
plot_pc took 1.40 seconds to run
plot_pc_drug took 2.87 seconds to run
Animal 020
plot_pc took 1.20 seconds to run
plot_pc took 1.18 seconds to run
plot_pc_drug took 2.40 seconds to run
Animal 021
plot_pc took 1.11 seconds to run
plot_pc took 1.10 seconds to run
plot_pc_drug took 2.23 seconds to run
Animal 022
plot_pc took 1.08 seconds to run
plot_pc took 1.07 seconds to run
plot_pc_drug took 2.17 seconds to run
Animal 023
plot_pc took 1.05 seconds to run
plot_pc took 1.08 seconds to run
plot_pc_drug took 2.14 seconds to run
Animal 024
plot_pc took 1.01 seconds to run
plot_pc took 1.02 seconds to run
plot_pc_drug took 2.12 seconds to run
Animal 025
plot_pc took 1.02 seconds to run
plot_pc took 1.04 seconds to run
plot_pc_drug took 2.08 seconds to run

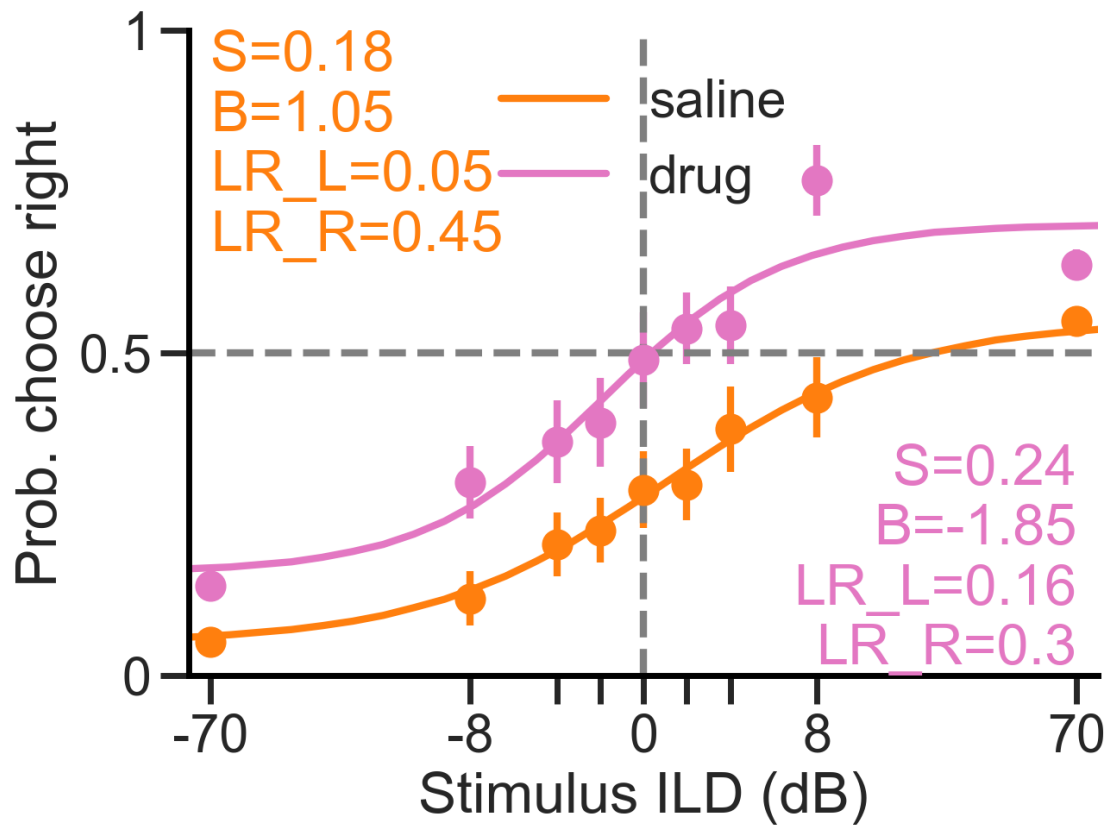


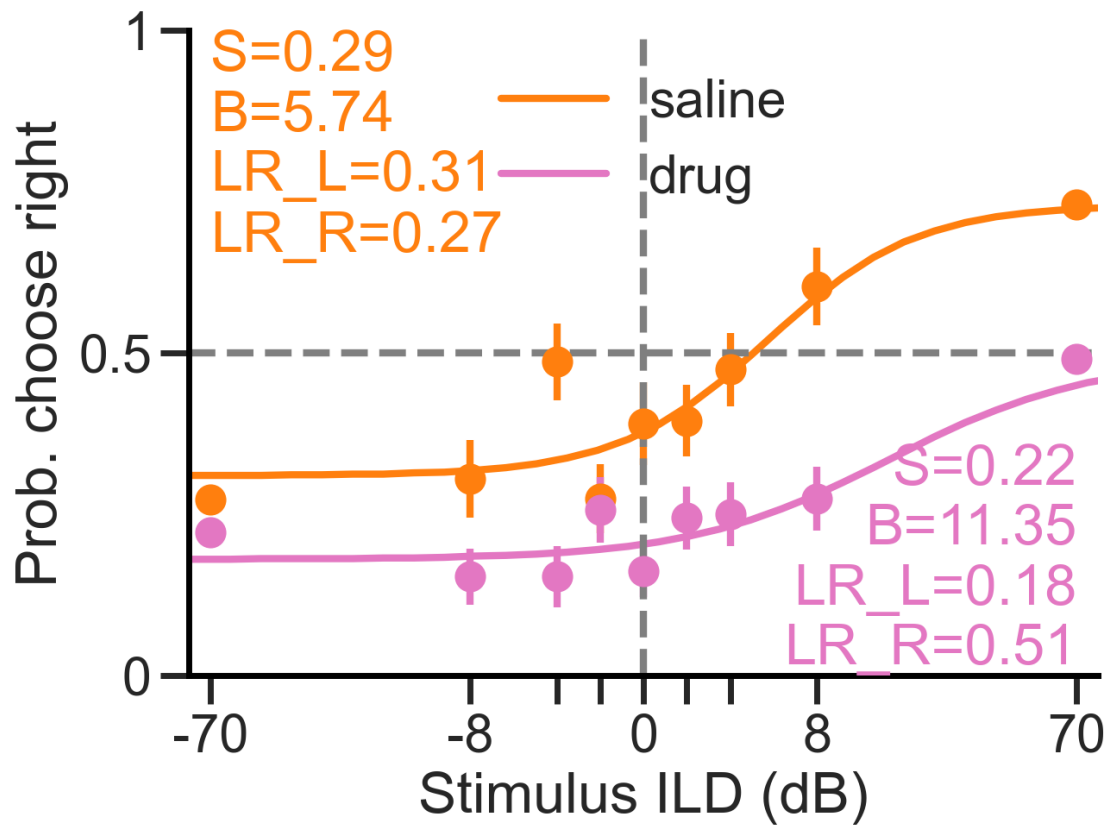


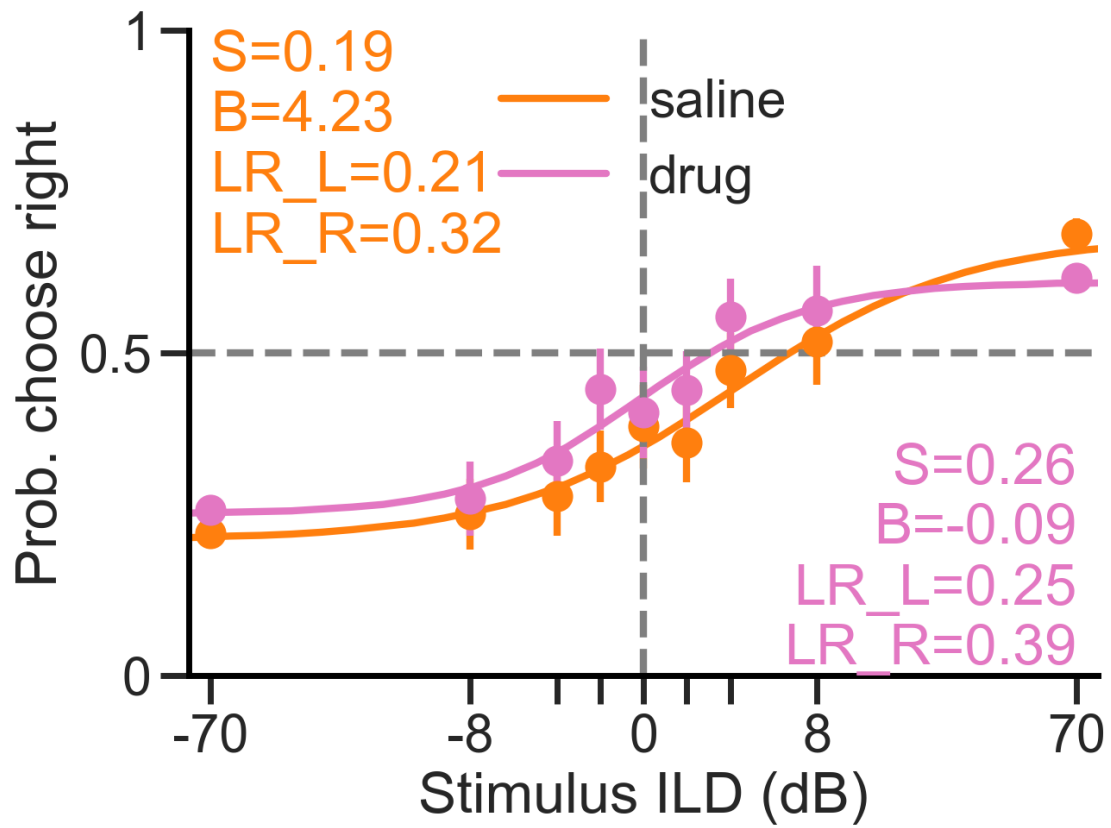


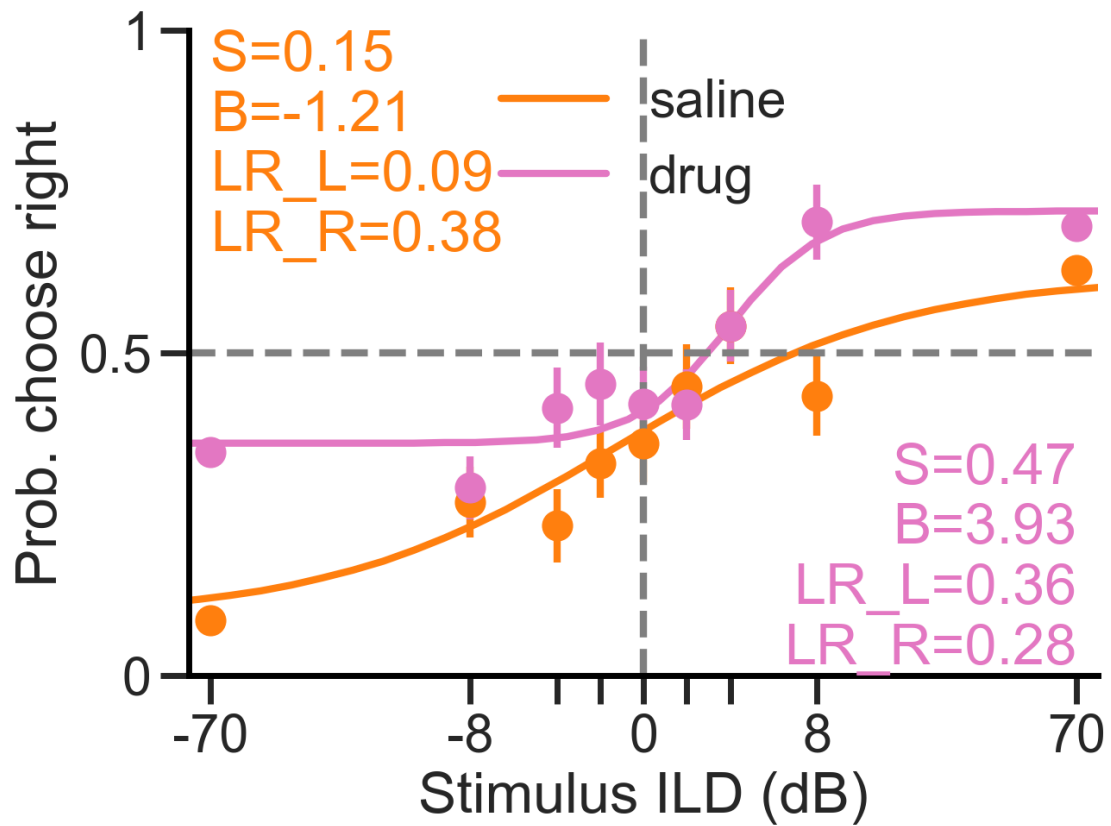






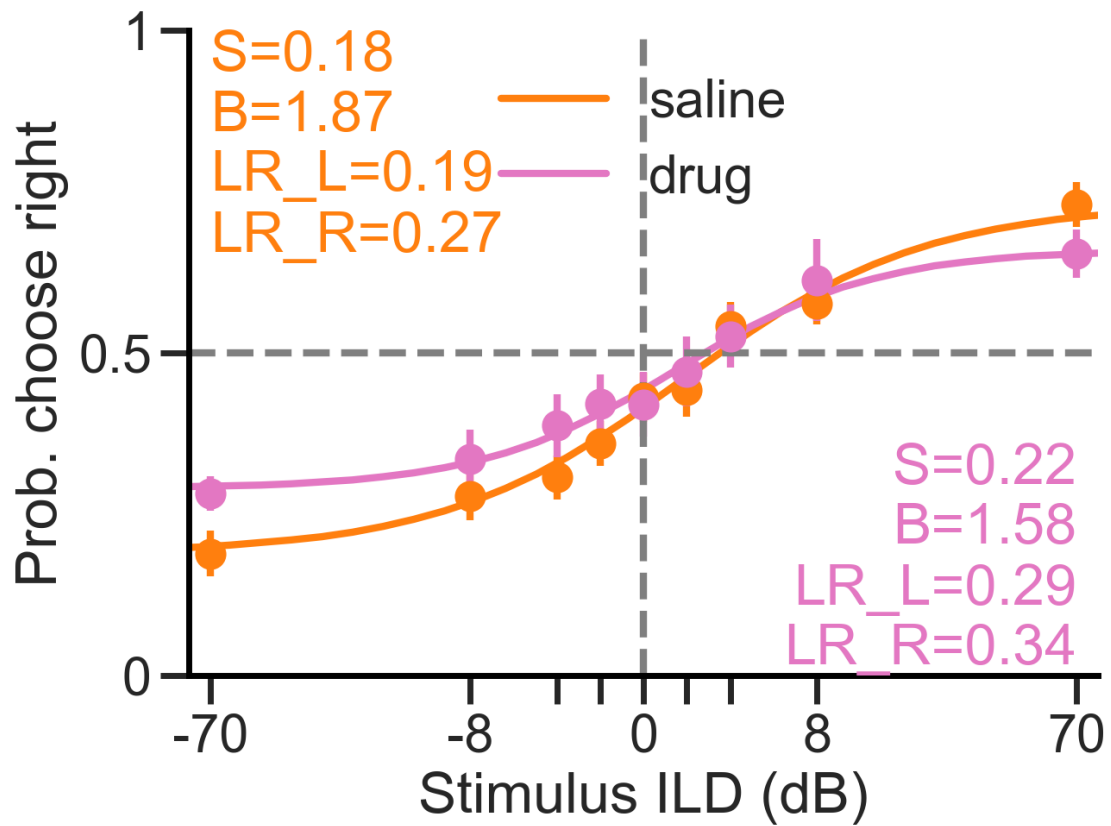






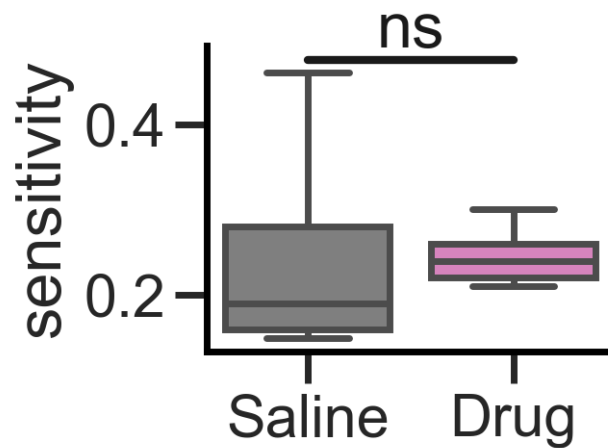
Mean across animals

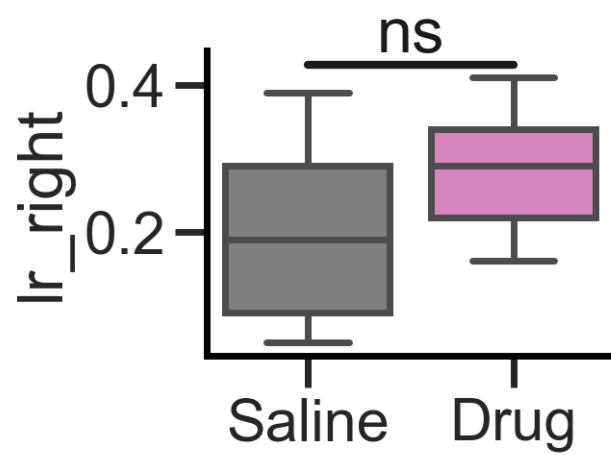
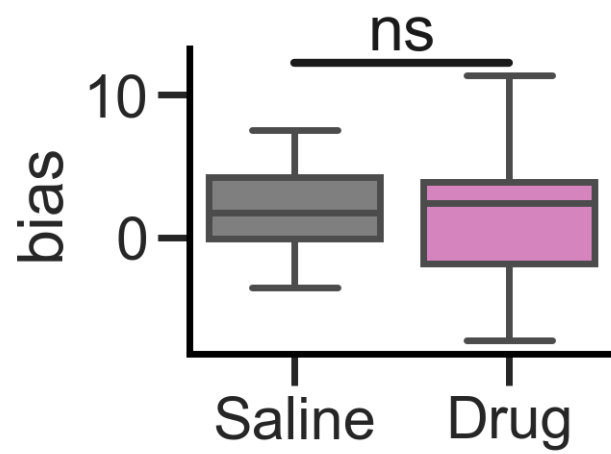
plot_mean_pc took 9.86 seconds to run
 plot_mean_pc took 9.76 seconds to run
 plot_mean_pc_drug took 19.64 seconds to run

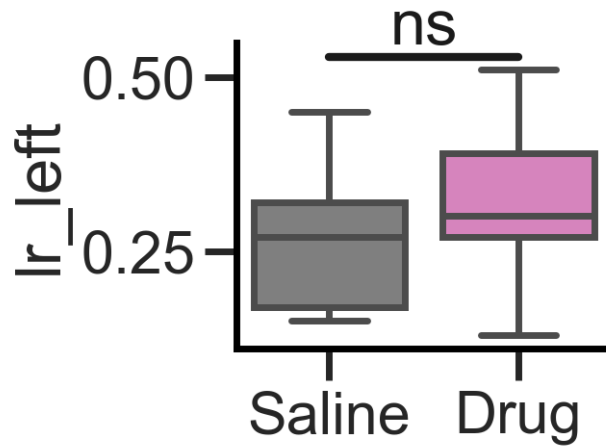


Stats

sensitivity: t-statistic = -0.561, p-value = 0.583
 bias: t-statistic = 0.111, p-value = 0.913
 lr_right: t-statistic = -1.813, p-value = 0.089
 lr_left: t-statistic = -1.068, p-value = 0.301







1.2.2 Probability choose REPEAT

Single animals

Animal 014

plot_pc took 1.13 seconds to run

plot_pc took 1.13 seconds to run

plot_pc_drug took 2.29 seconds to run

Animal 016

plot_pc took 1.24 seconds to run

plot_pc took 1.25 seconds to run

plot_pc_drug took 2.51 seconds to run

Animal 017

plot_pc took 1.30 seconds to run

plot_pc took 1.34 seconds to run

plot_pc_drug took 2.66 seconds to run

Animal 020

plot_pc took 1.09 seconds to run

plot_pc took 1.09 seconds to run

plot_pc_drug took 2.20 seconds to run

Animal 021

plot_pc took 1.07 seconds to run

plot_pc took 1.06 seconds to run

plot_pc_drug took 2.24 seconds to run

Animal 022

plot_pc took 1.05 seconds to run

plot_pc took 1.04 seconds to run

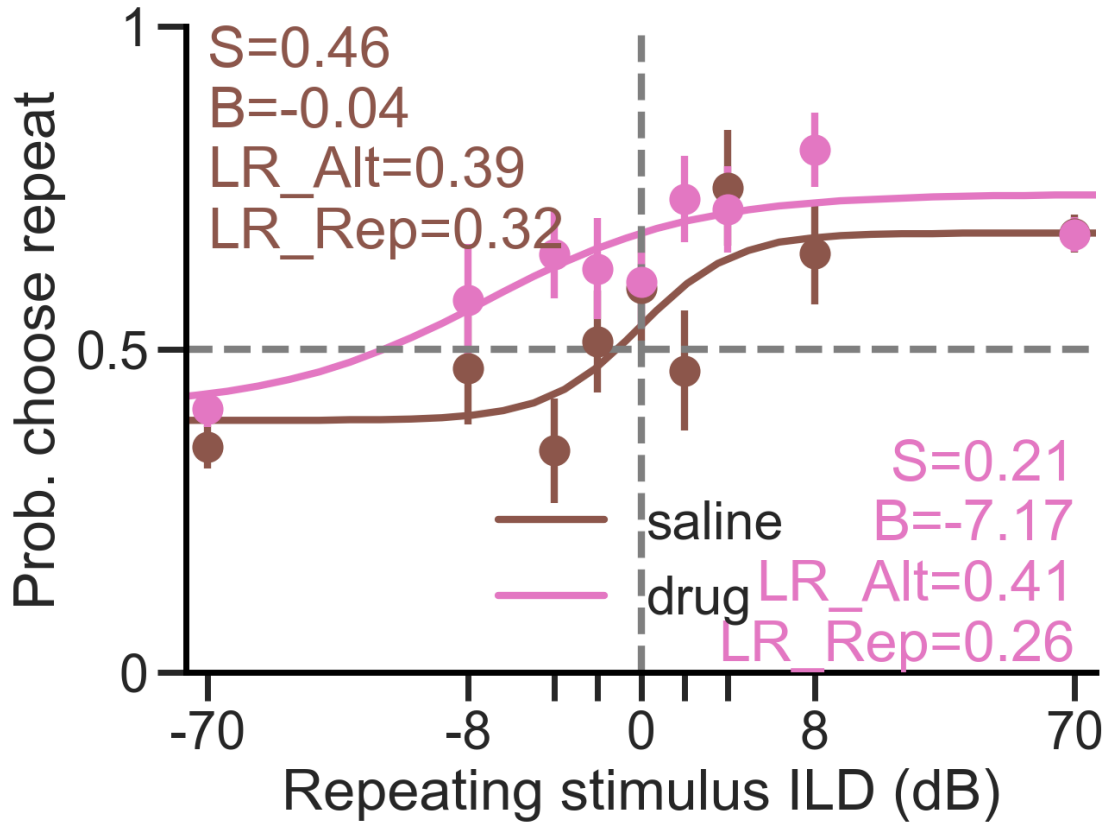
plot_pc_drug took 2.11 seconds to run

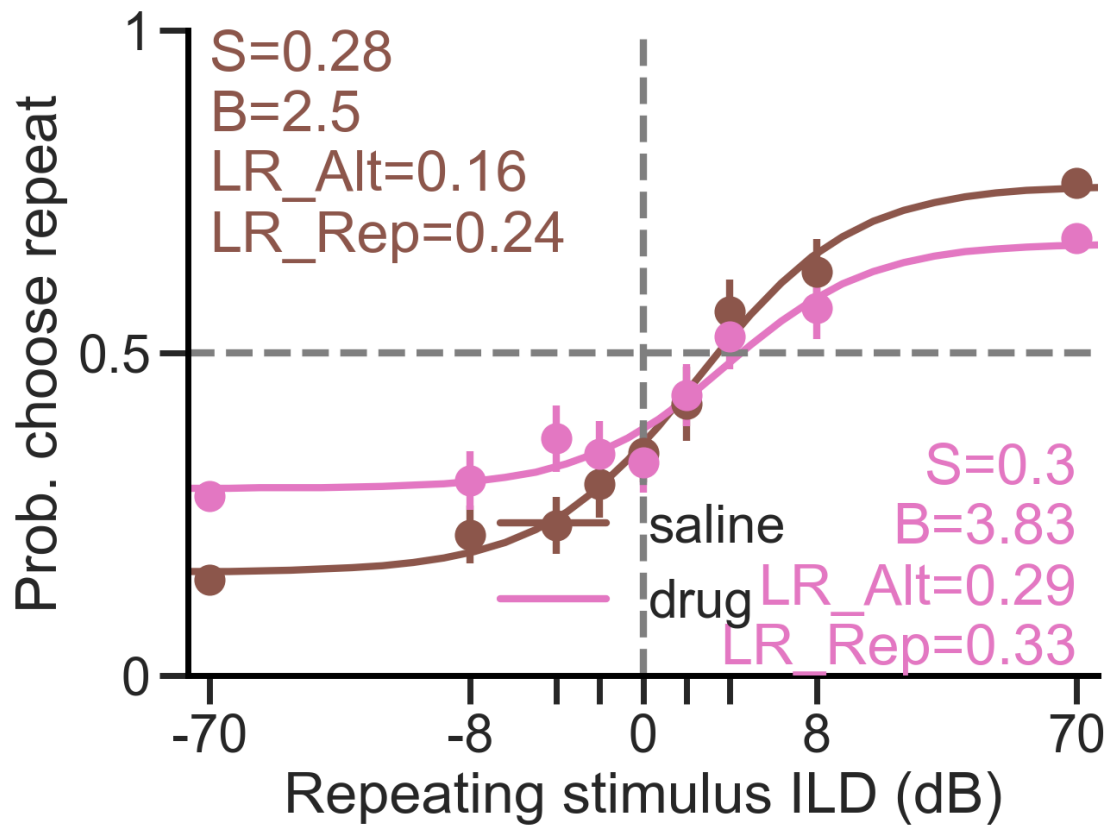
Animal 023

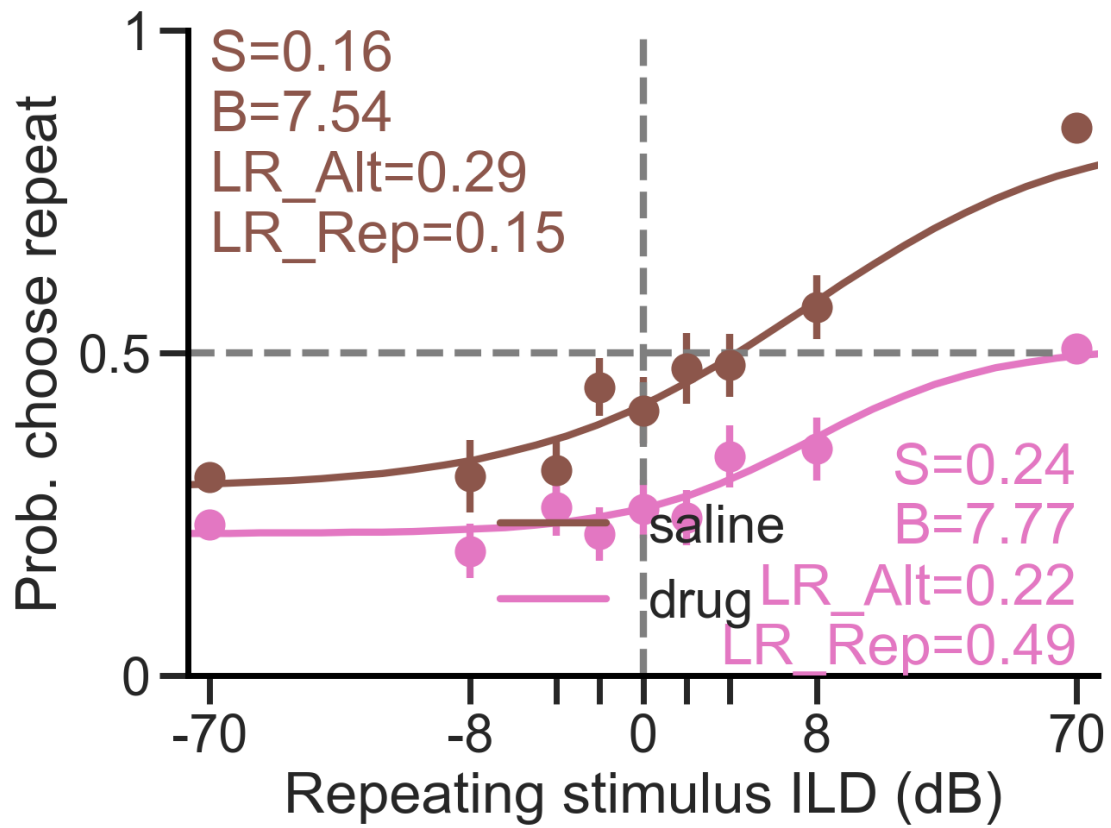
plot_pc took 1.05 seconds to run

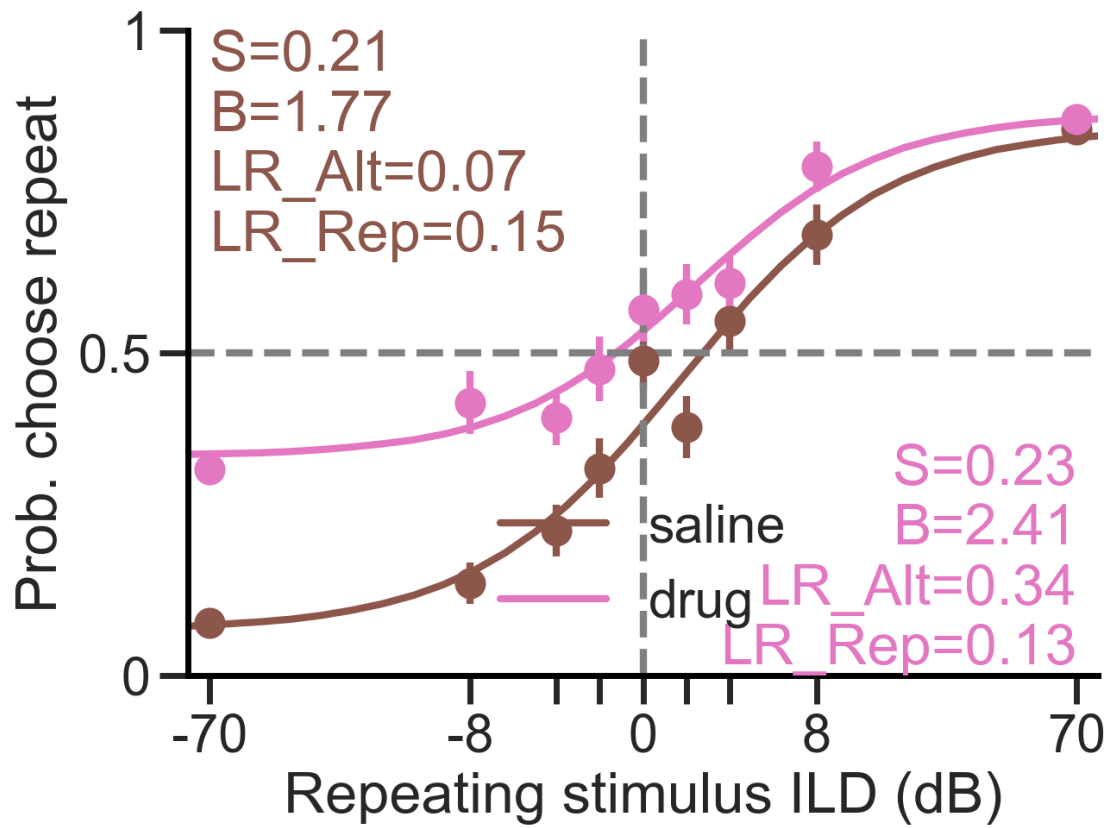
plot_pc took 1.01 seconds to run

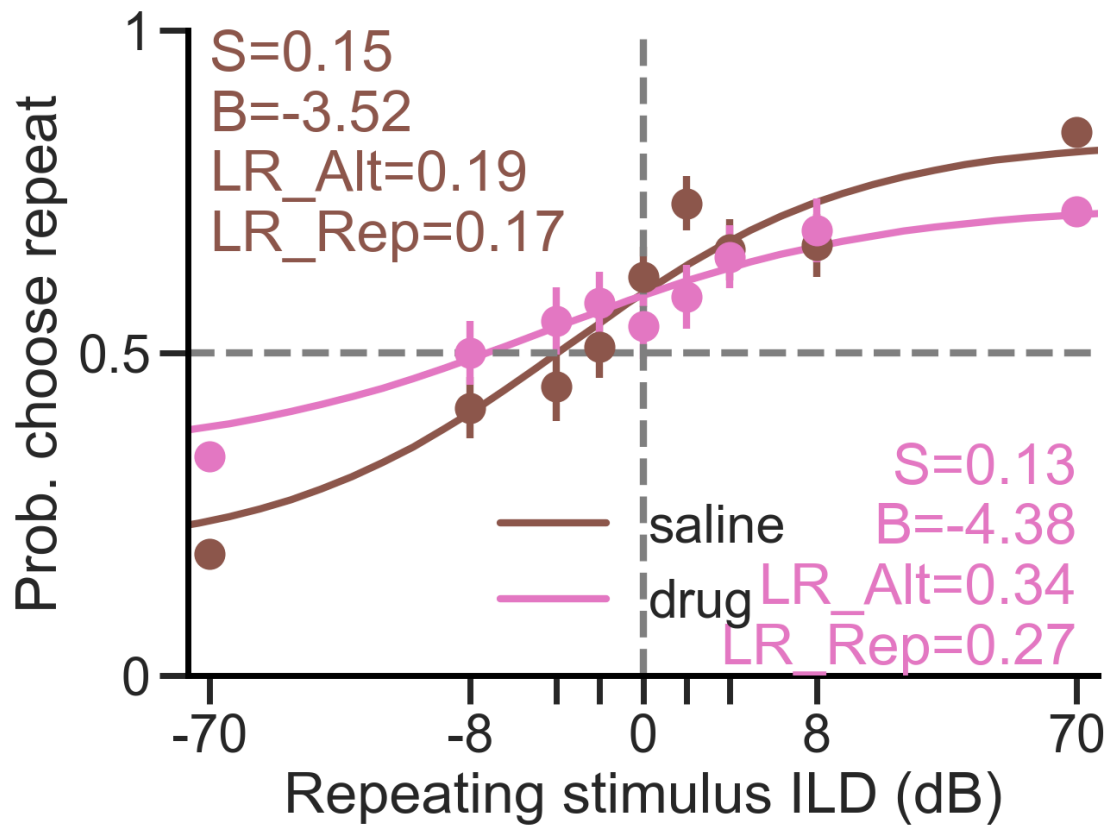
plot_pc_drug took 2.08 seconds to run
 Animal 024
 plot_pc took 1.06 seconds to run
 plot_pc took 1.05 seconds to run
 plot_pc_drug took 2.13 seconds to run
 Animal 025
 plot_pc took 0.98 seconds to run
 plot_pc took 1.02 seconds to run
 plot_pc_drug took 2.03 seconds to run

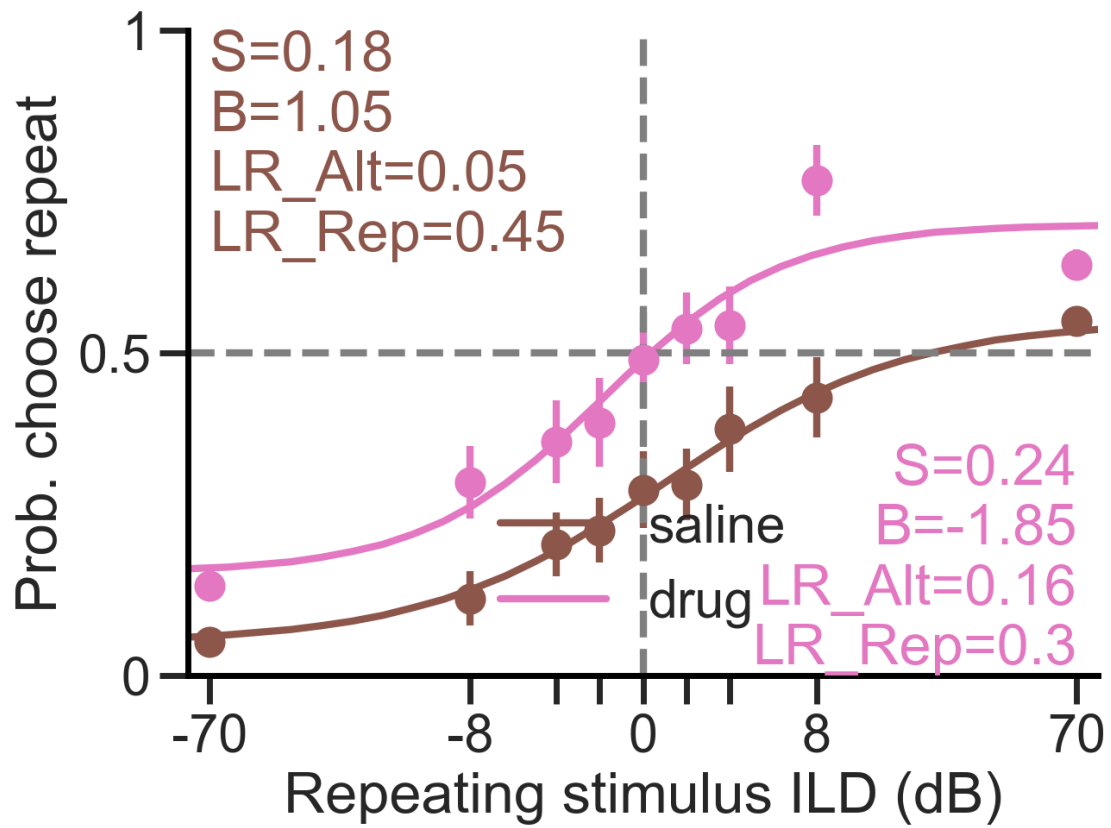


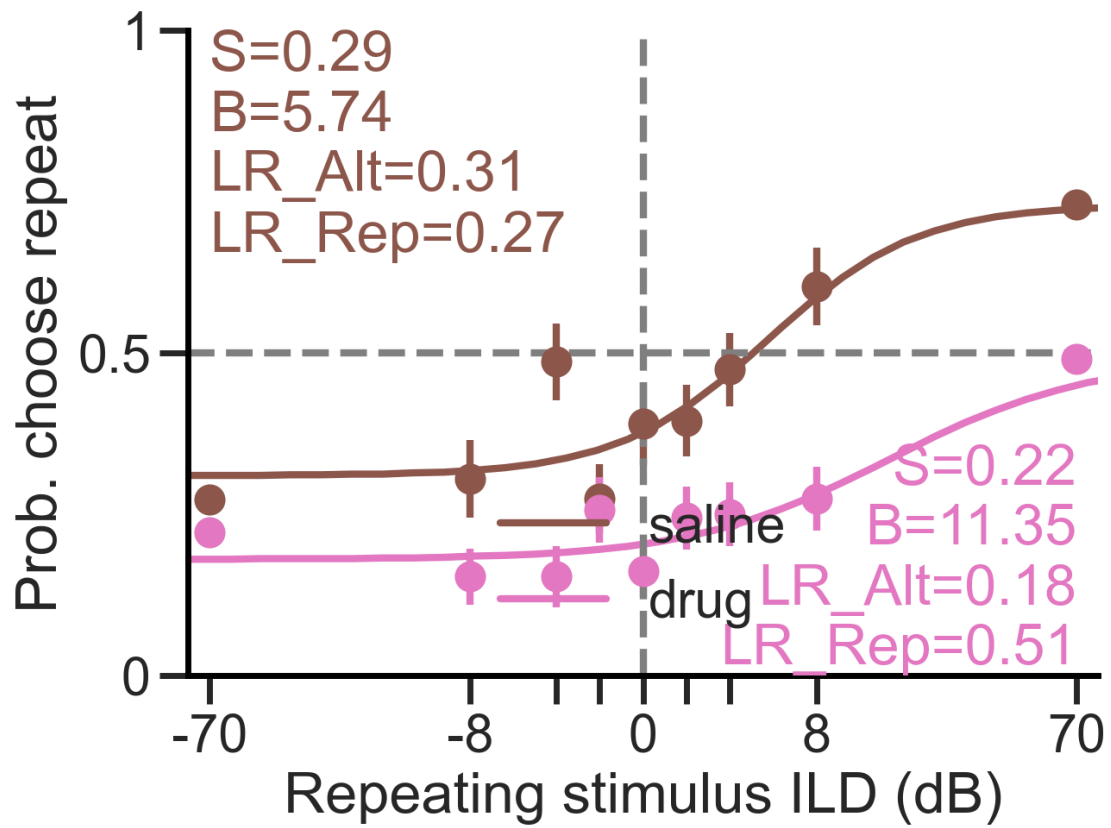


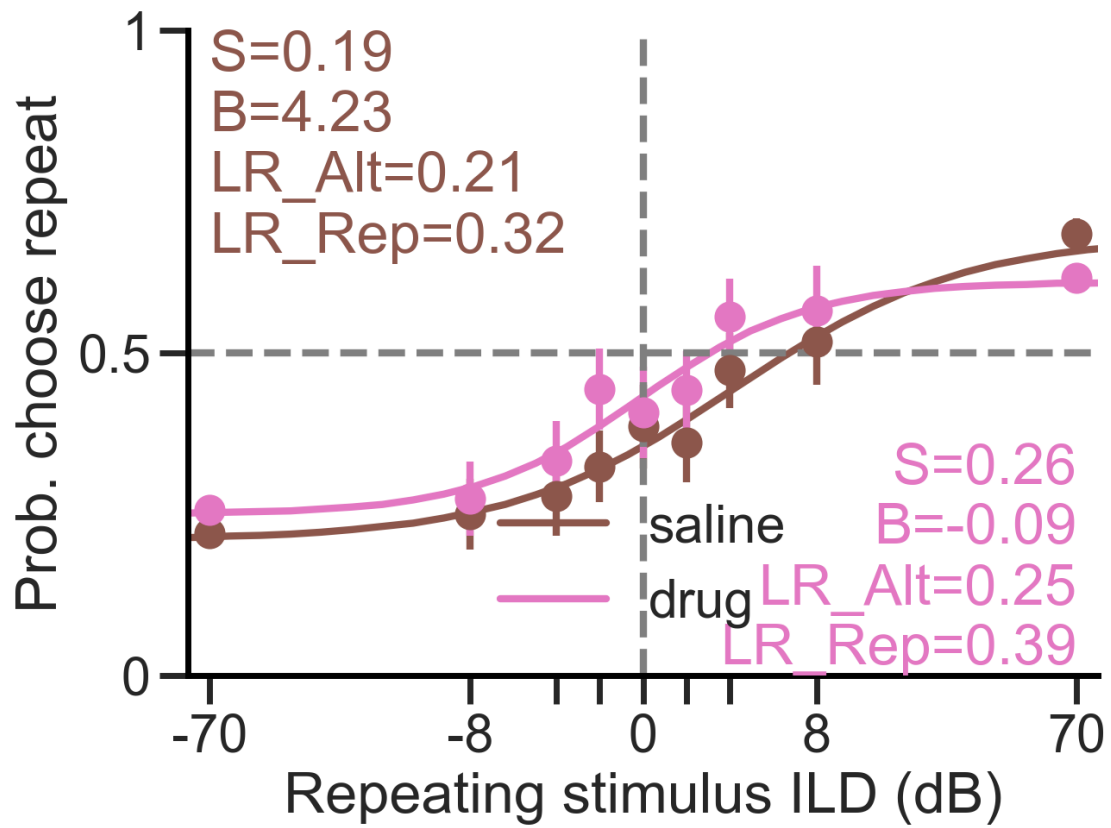


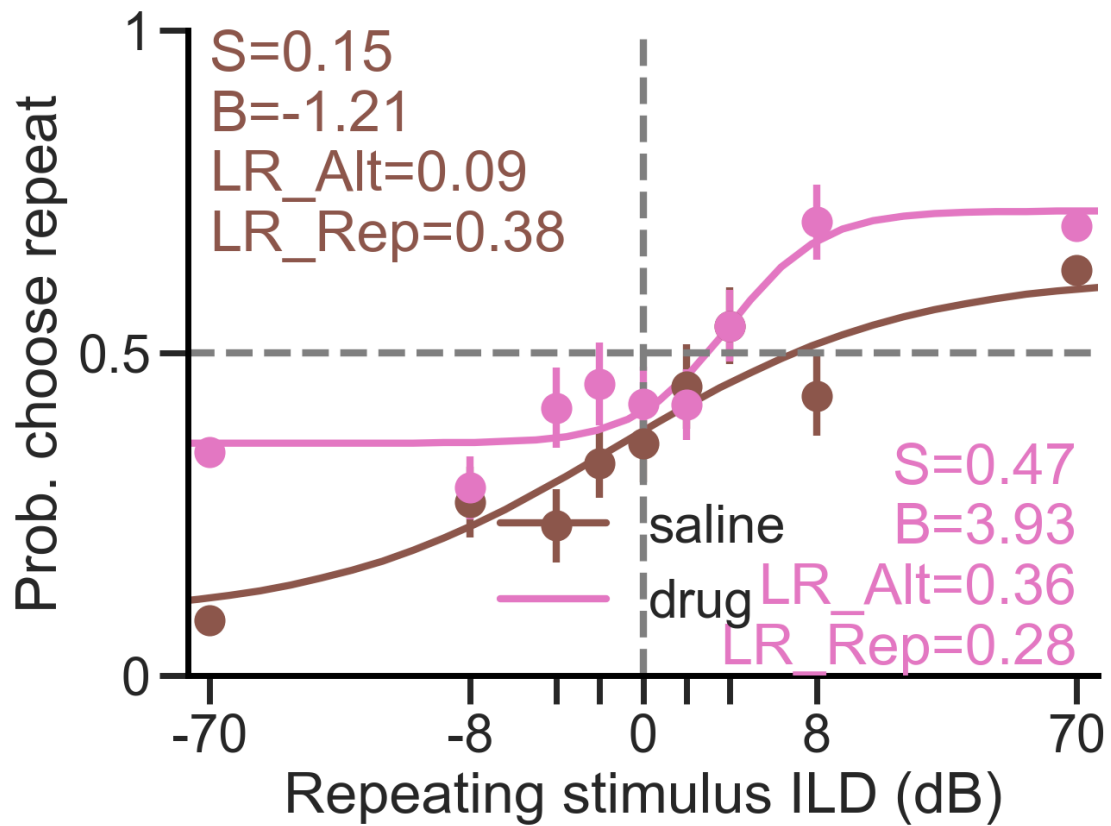






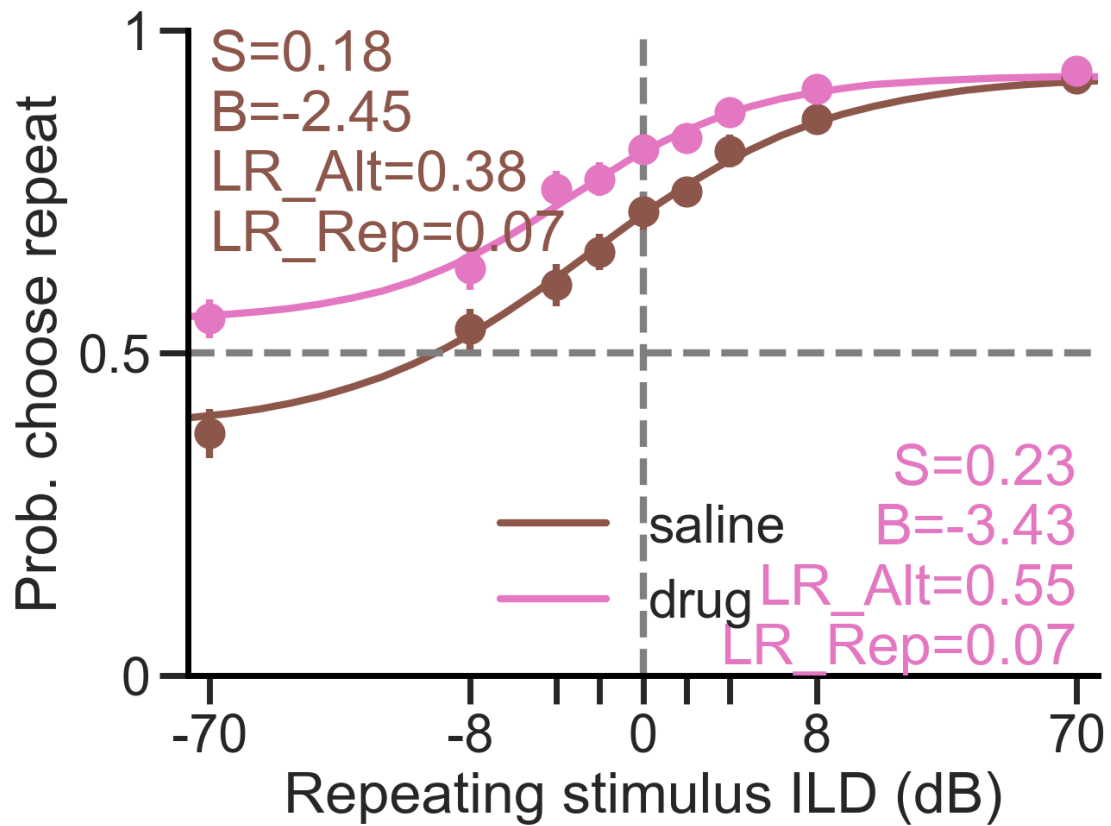






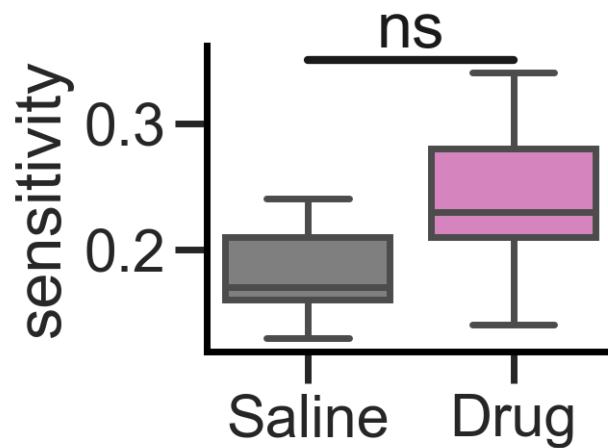
Mean across animals

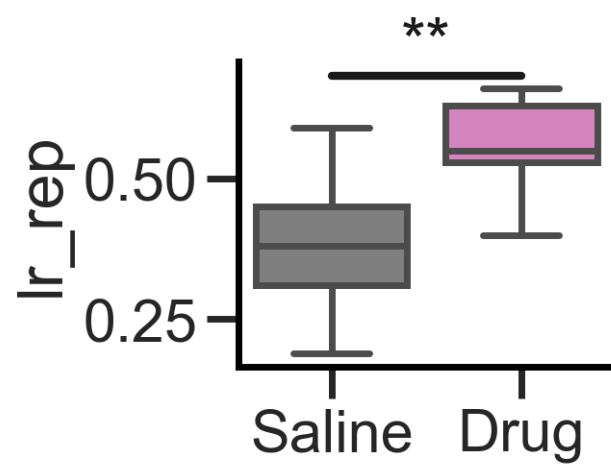
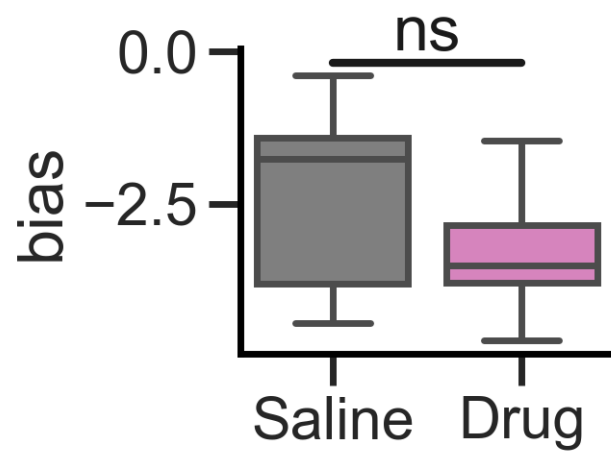
plot_mean_pc took 10.12 seconds to run
 plot_mean_pc took 10.12 seconds to run
 plot_mean_pc_drug took 20.26 seconds to run

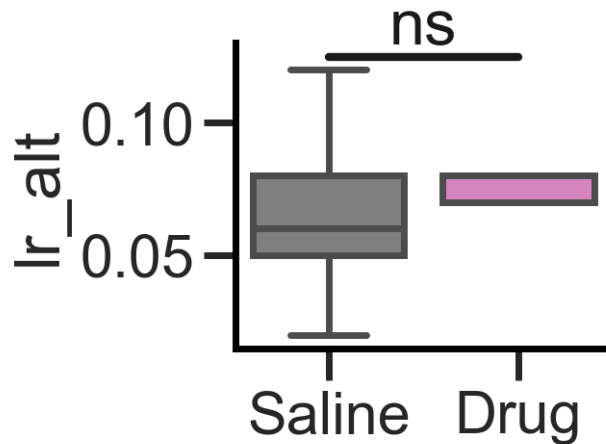


Stats

sensitivity: t-statistic = -0.249, p-value = 0.806
 bias: t-statistic = 1.293, p-value = 0.214
 lr_rep: t-statistic = -3.346, p-value = 0.004
 lr_alt: t-statistic = 0.307, p-value = 0.763







1.3 Kernels

1.3.1 Pshycophysical kernels

Mean across animals

2AFC_6

C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6

Getting kernel of animal 014 (1/9)

get_pk took 10.72 seconds to run

Getting kernel of animal 016 (2/9)

get_pk took 17.49 seconds to run

Getting kernel of animal 017 (3/9)

get_pk took 16.75 seconds to run

Getting kernel of animal 020 (4/9)

get_pk took 29.29 seconds to run

Getting kernel of animal 021 (5/9)

get_pk took 15.04 seconds to run

Getting kernel of animal 022 (6/9)

get_pk took 9.73 seconds to run

Getting kernel of animal 023 (7/9)

get_pk took 9.64 seconds to run

Getting kernel of animal 024 (8/9)

get_pk took 12.46 seconds to run

Getting kernel of animal 025 (9/9)

get_pk took 15.39 seconds to run

get_mean_pk took 136.72 seconds to run

2AFC_6

C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6

Getting kernel of animal 014 (1/9)

get_pk took 12.63 seconds to run

Getting kernel of animal 016 (2/9)

```

get_pk took 15.06 seconds to run
Getting kernel of animal 017 (3/9)
get_pk took 14.42 seconds to run
Getting kernel of animal 020 (4/9)
get_pk took 22.81 seconds to run
Getting kernel of animal 021 (5/9)
get_pk took 14.24 seconds to run
Getting kernel of animal 022 (6/9)
get_pk took 14.44 seconds to run
Getting kernel of animal 023 (7/9)
get_pk took 12.21 seconds to run
Getting kernel of animal 024 (8/9)
get_pk took 12.79 seconds to run
Getting kernel of animal 025 (9/9)
get_pk took 16.07 seconds to run

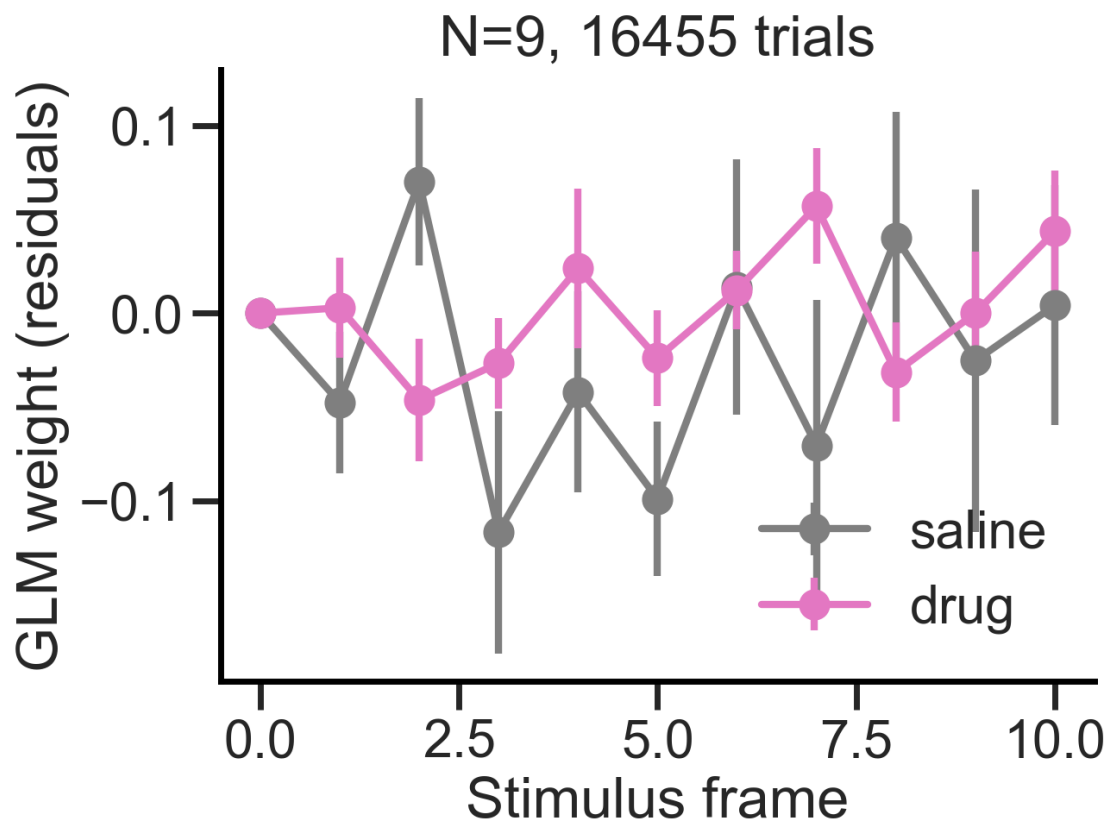
```

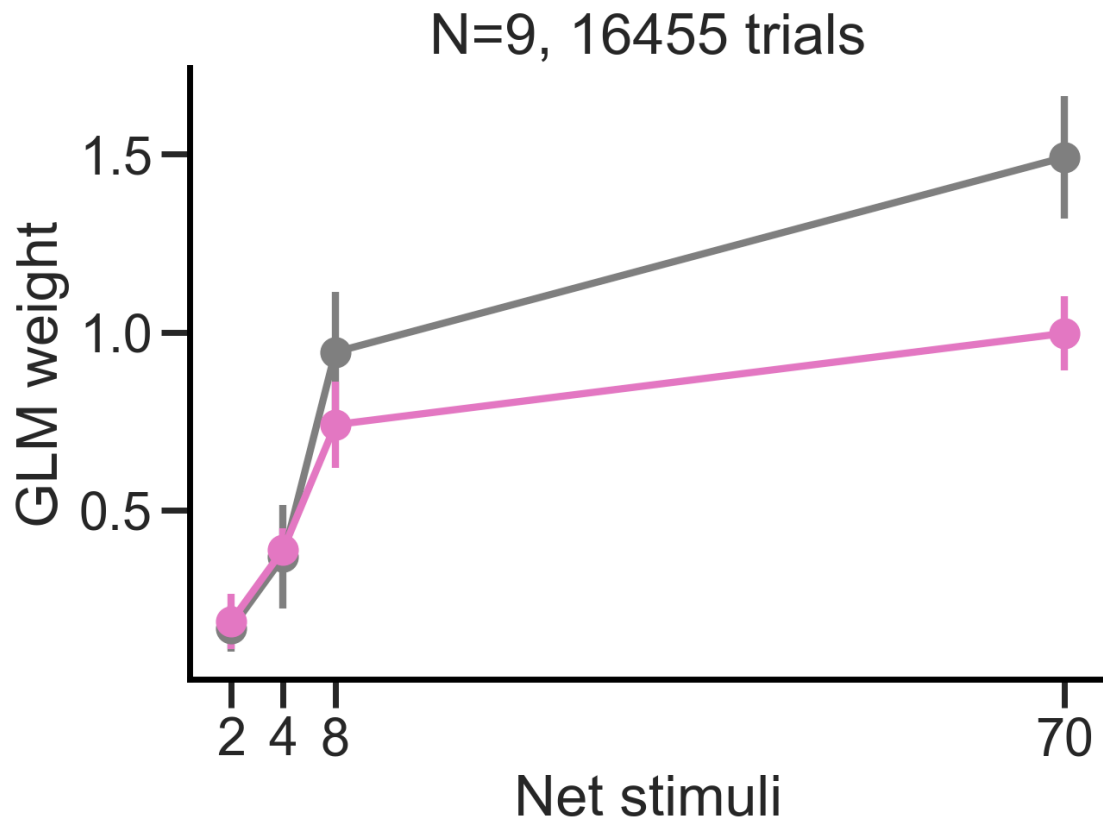
No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

```

get_mean_pk took 135.15 seconds to run
plot_drug_pk took 271.89 seconds to run

```





1.3.2 History kernels

1.4 Serial dependence

0.70% of premature lick trials (before response window)

Port1In or Port2In are already lists

	Trial	Side	PreviousSide	RepSide	RepTrial	Reward	Punish	Miss	\
0	1	1	1.0	1	NaN	1	0	0	
1	2	1	1.0	1	1.0	1	0	0	
2	3	0	1.0	0	0.0	1	0	0	
3	4	0	0.0	1	1.0	1	0	0	
4	5	1	0.0	0	0.0	1	0	0	

	WrongLick	Hit	...	SessionEnd	\
0	0	1.0	...	12:19:39.607264	
1	0	1.0	...	12:19:39.607264	
2	0	1.0	...	12:19:39.607264	
3	0	1.0	...	12:19:39.607264	
4	0	1.0	...	12:19:39.607264	

```

                                LicksLeft  \
0                                []
1                                []
2  [1.8217999999999996, 1.9610000000000003, 2.059...
3  [1.8061999999999996, 1.9127999999999998, 2.009...
4                                []

                                LicksRight  nLicksLeft  nLicksRight  \
0          [1.8024, 1.9372000000000003, 2.0817]          0          3
1  [1.9699000000000004, 2.1862, 2.2955, 2.4038999...          0          8
2          [2.5639000000000003, 2.6929]          8          2
3          [2.516, 2.6639999999999997]          8          2
4  [1.7715, 1.9083999999999999, 1.9987, 2.1058999...          0          9

                                nLicks                                ILI_Left  \
0 -3.639988                                NaN
1  1.360012                                NaN
2  1.360012  [0.13920000000000066, 0.0987999999999978, 0.1...
3  1.360012  [0.10660000000000025, 0.0968, 0.111600000000000...
4  2.360012                                NaN

                                ILI_Right          RT          RT2
0          [0.13480000000000025, 0.14449999999999985] -0.026308 -0.026308
1  [0.21629999999999995, 0.10930000000000017, 0.10... 0.141192 0.141192
2          [0.12899999999999956] -0.006908 -0.006908
3          [0.1479999999999997] -0.022508 -0.022508
4  [0.13689999999999998, 0.09030000000000005, 0.10... -0.057208 -0.057208

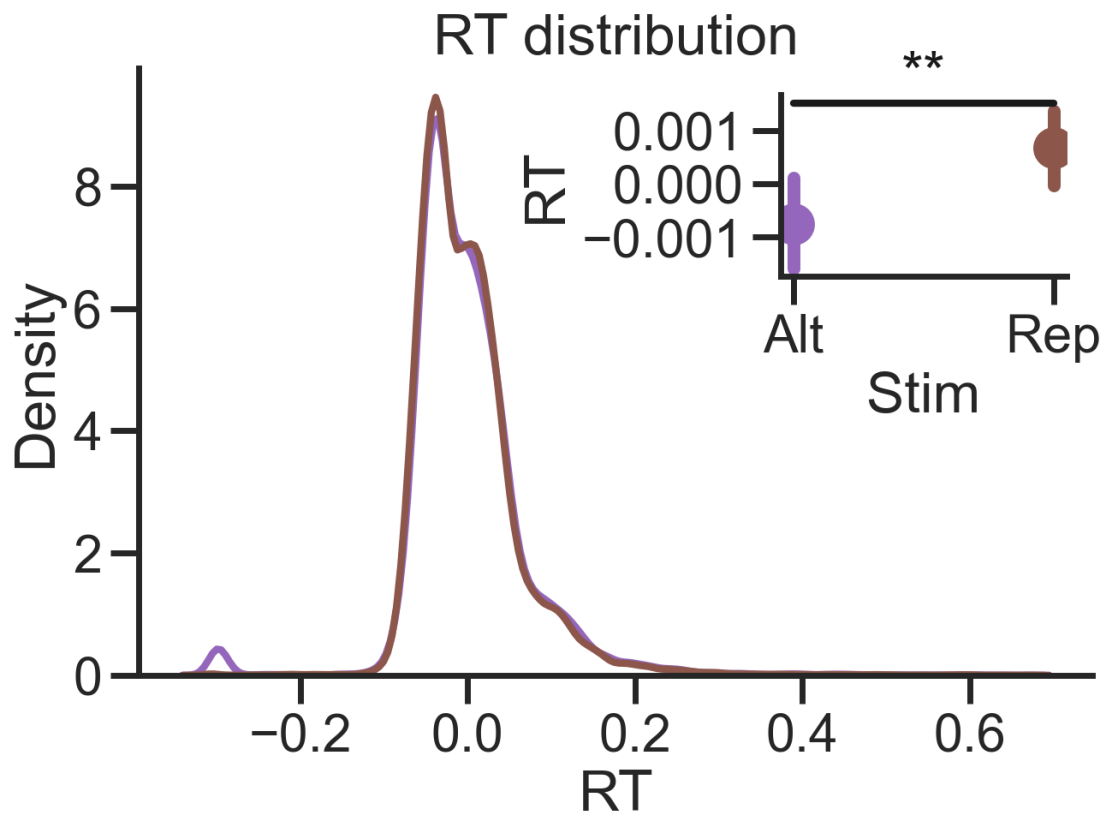
[5 rows x 93 columns]

array([-140., -78., -74., -72., -70., -68., -66., -62., -16.,
       -12., -10., -8., -6., -4., -2., 0., 2., 4.,
        6., 8., 10., 12., 16., 62., 66., 68., 70.,
        72., 74., 78., 140.])

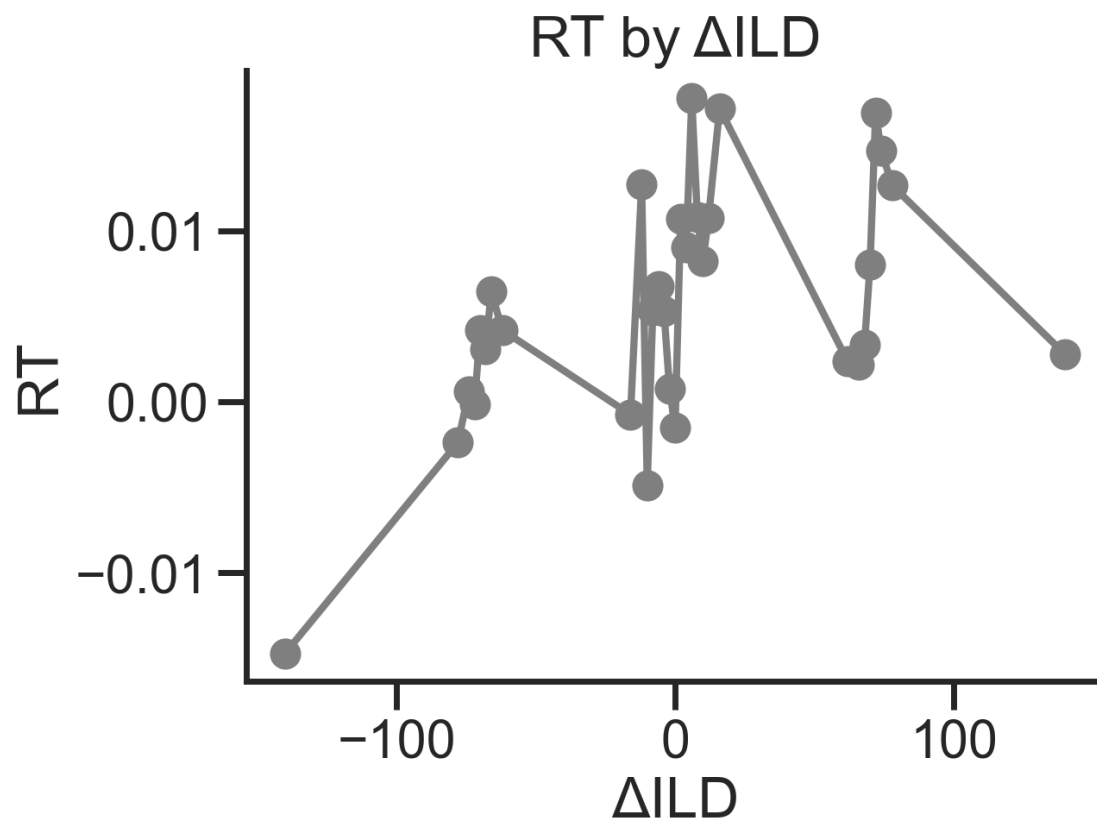
```

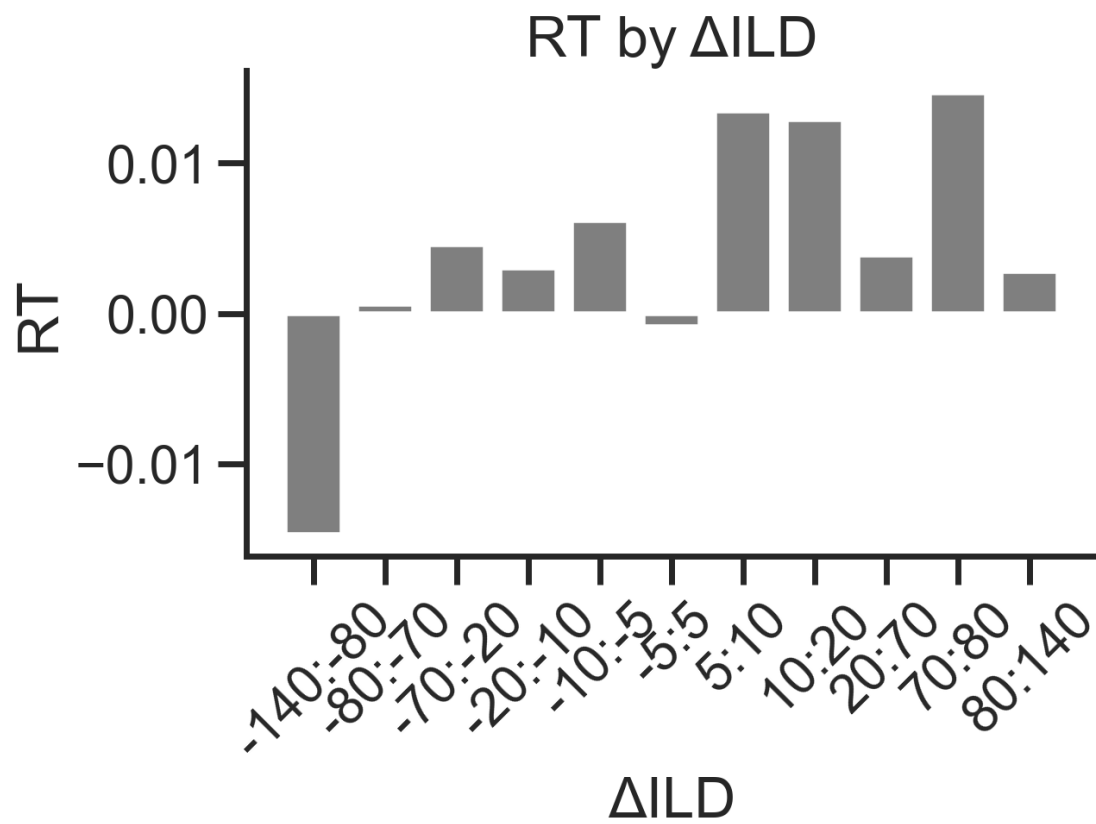
1.4.1 Measured as Reaction Times (RTs)

RT: t-statistic = -2.615, p-value = 0.009

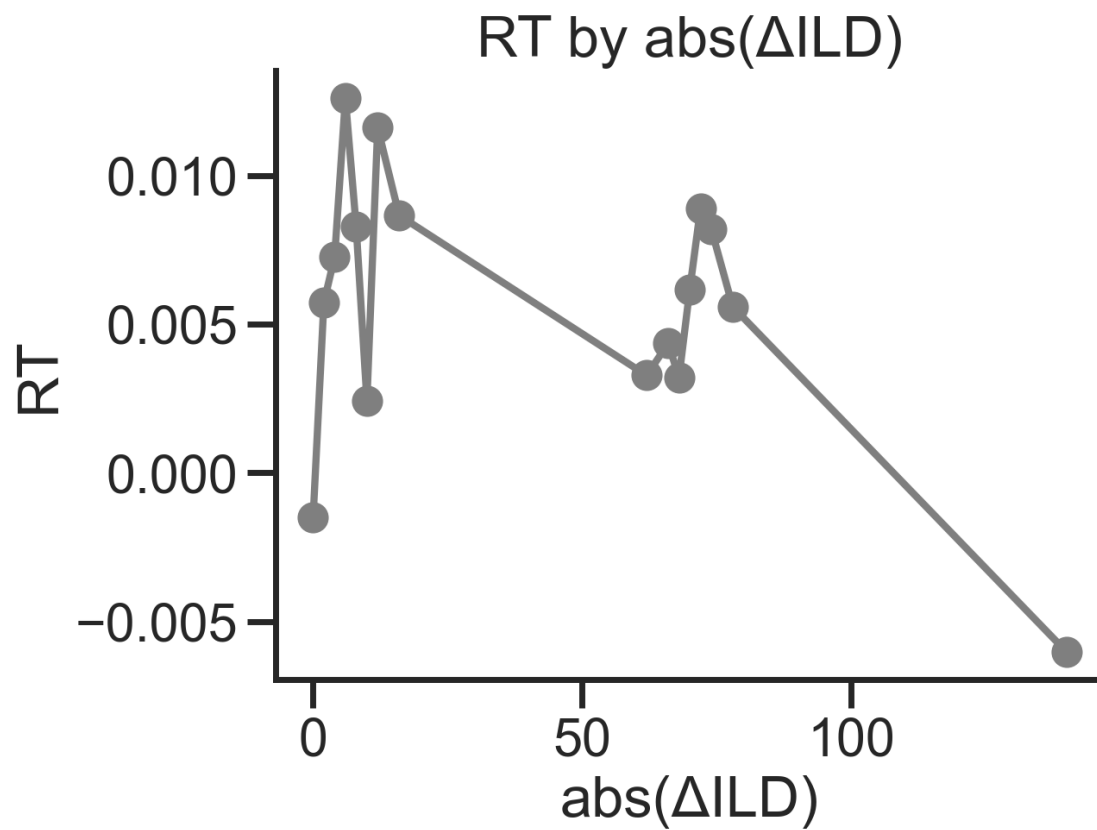


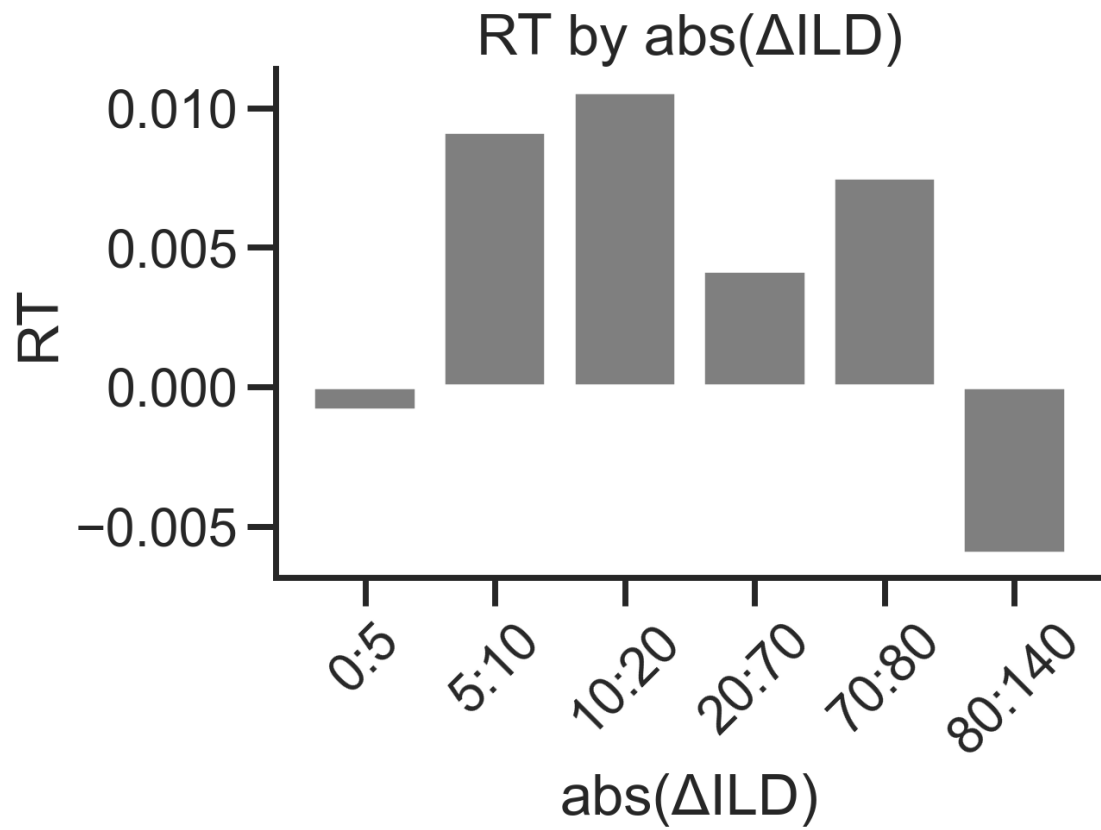
Δ ILD





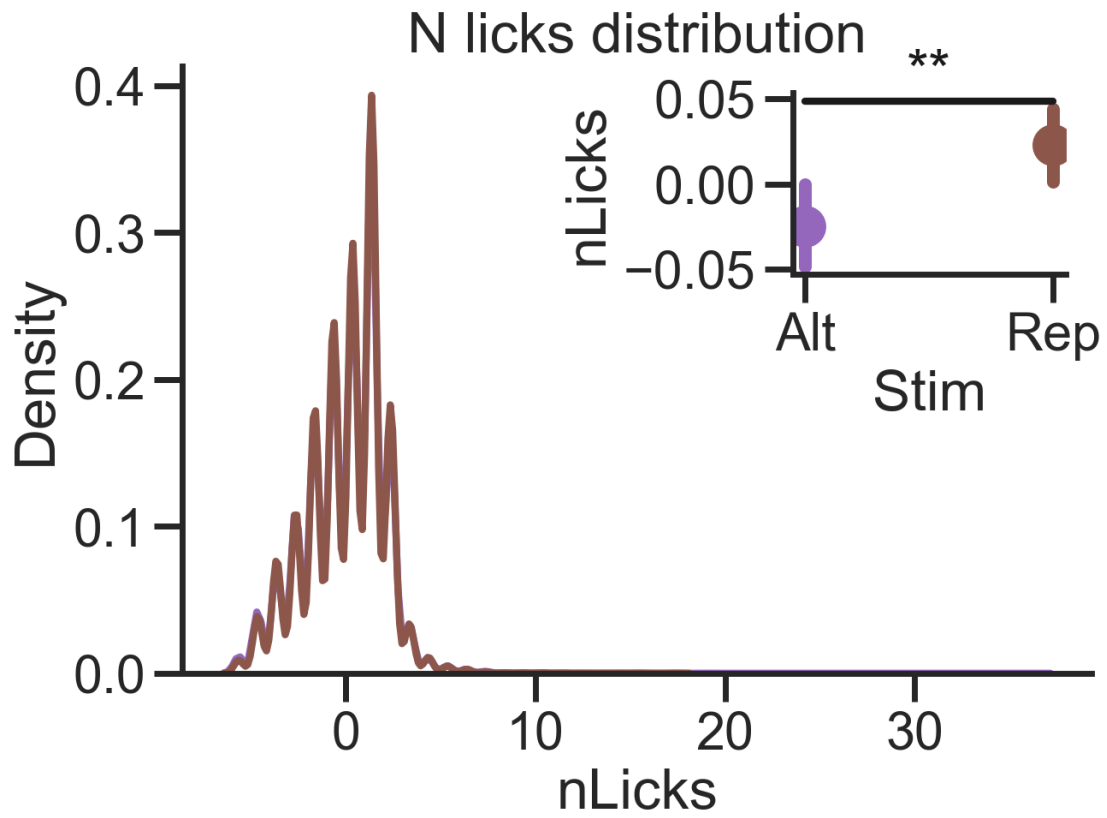
Absolute Δ ILD



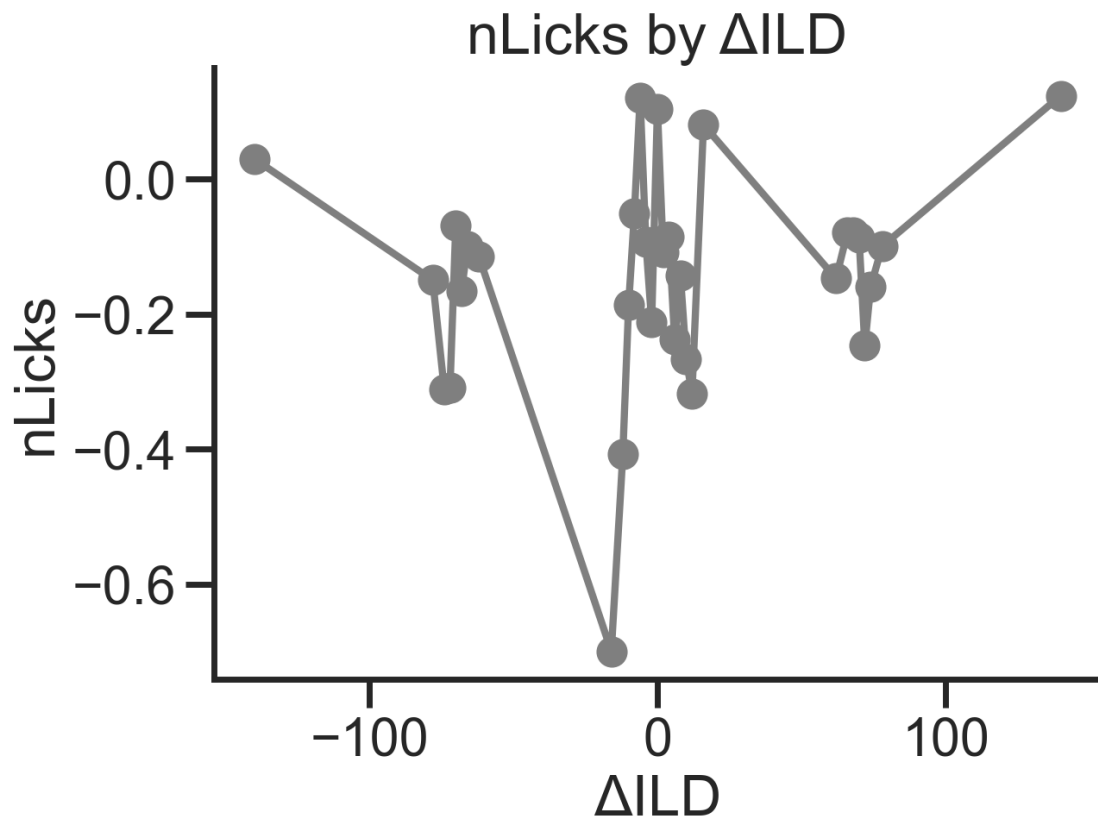


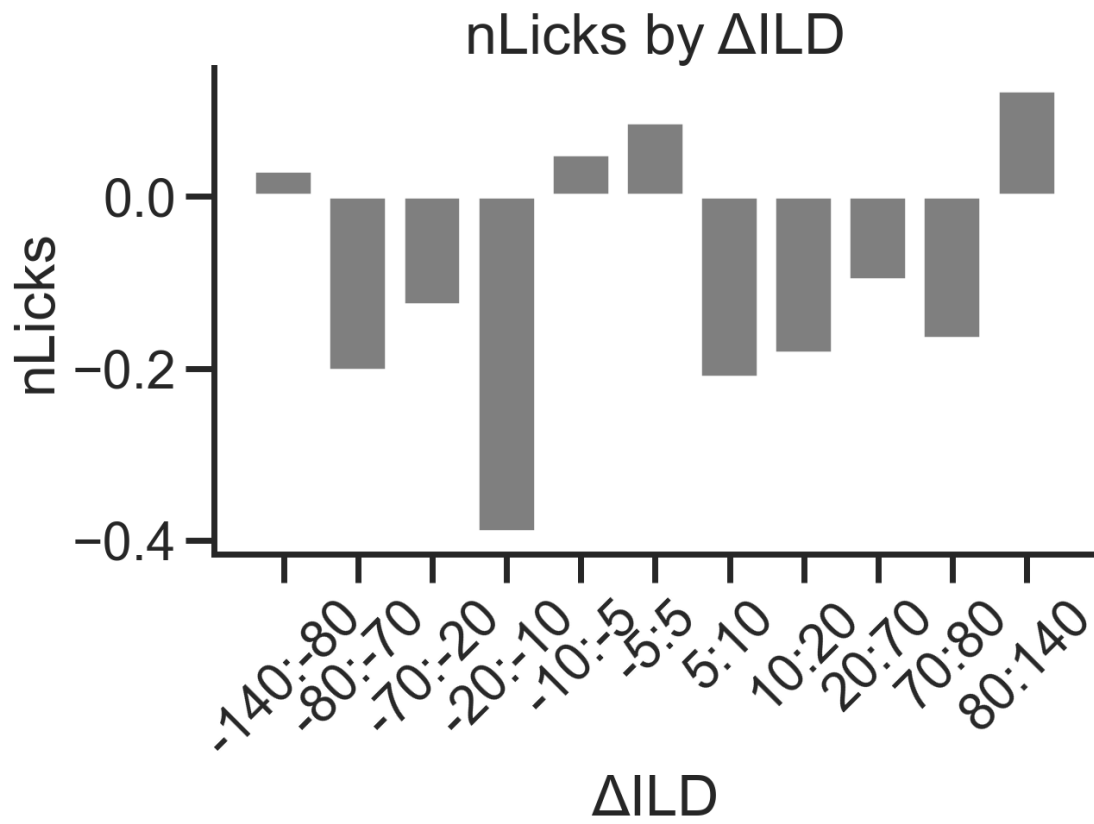
1.4.2 Measured as N licks

nLicks: t-statistic = -2.847, p-value = 0.004

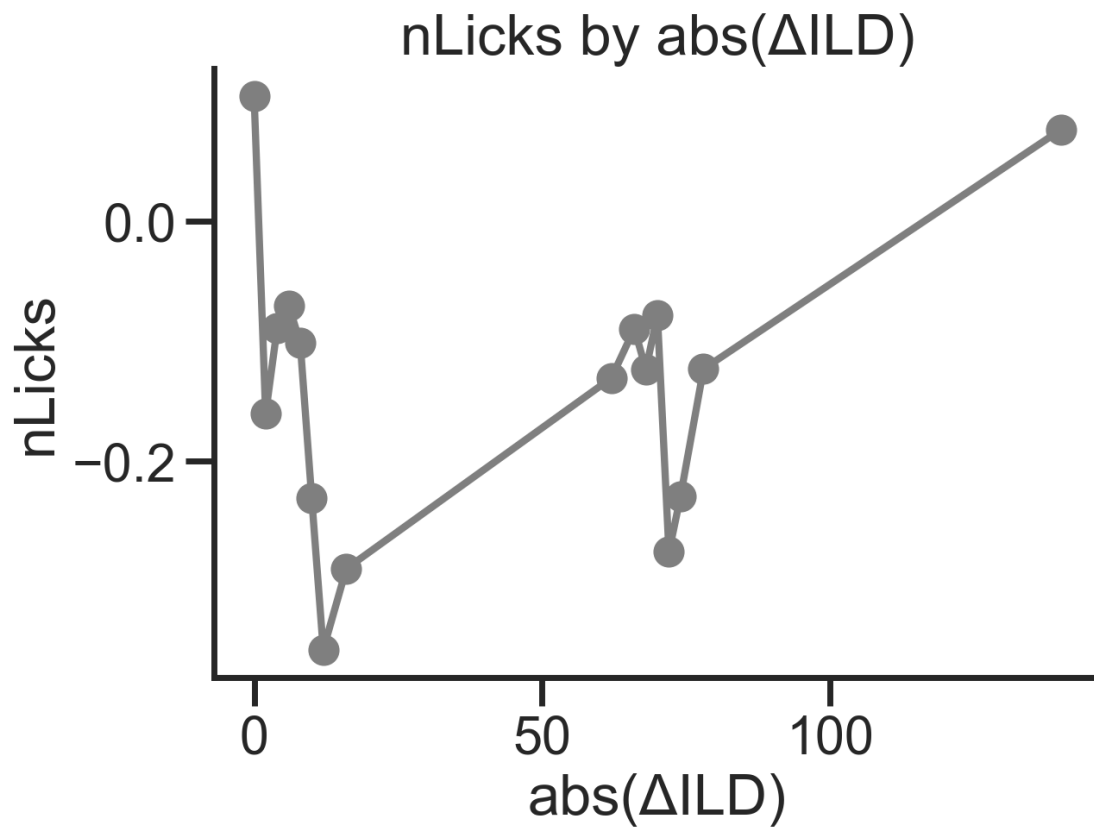


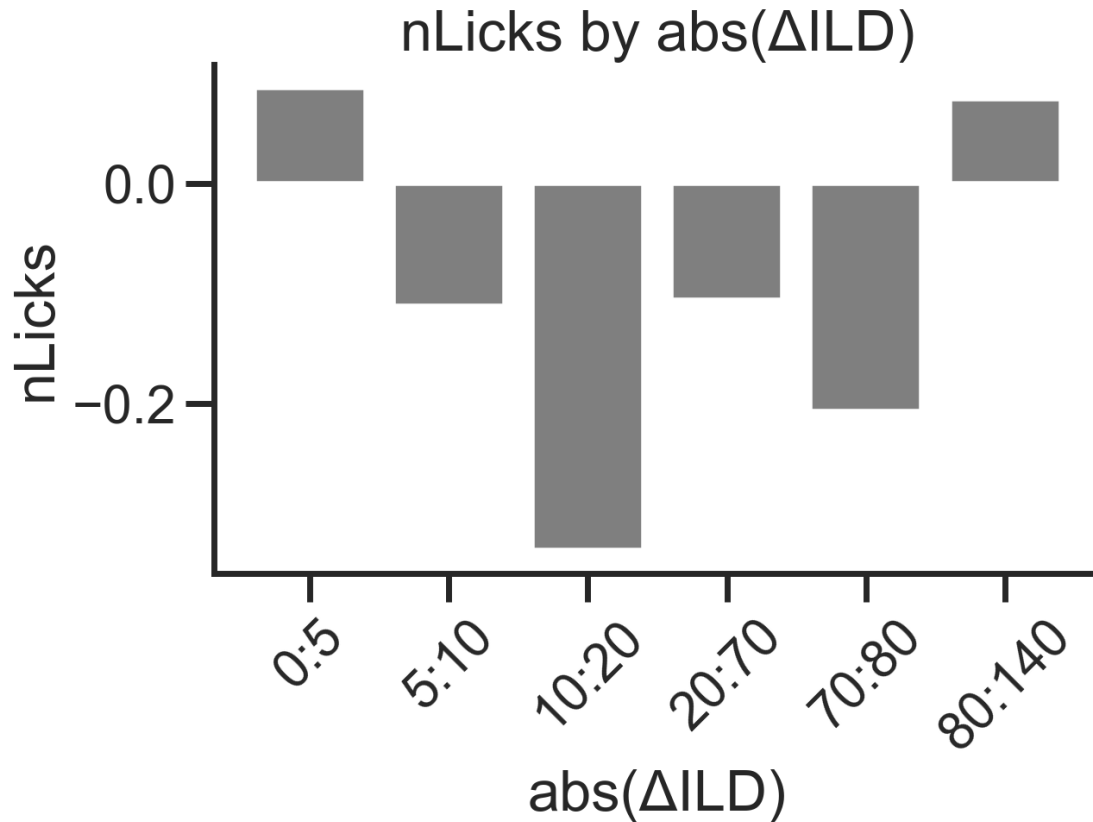
Δ ILD





Absolute Δ ILD





Save notebook's output

```
usage: jupyter-nbconvert [-h] [--debug] [--show-config] [--show-config-json]
                        [--generate-config] [-y] [--execute] [--allow-errors]
                        [--stdin] [--stdout] [--inplace] [--clear-output]
                        [--coalesce-streams] [--no-prompt] [--no-input]
                        [--allow-chromium-download]
                        [--disable-chromium-sandbox] [--show-input]
                        [--embed-images] [--sanitize-html]
                        [--log-level NbConvertApp.log_level]
                        [--config NbConvertApp.config_file]
                        [--to NbConvertApp.export_format]
                        [--template TemplateExporter.template_name]
                        [--template-file TemplateExporter.template_file]
                        [--theme HTMLExporter.theme]
                        [--sanitize_html HTMLExporter.sanitize_html]
                        [--writer NbConvertApp.writer_class]
                        [--post NbConvertApp.postprocessor_class]
                        [--output NbConvertApp.output_base]
                        [--output-dir FilesWriter.build_directory]
```

```
        [--reveal-prefix SlidesExporter.reveal_url_prefix]
        [--nbformat NotebookExporter.nbformat_version]
        [extra_args ...]
jupyter-nbconvert: error: unrecognized arguments: # As HTML
```